

Pinning Impulsive Control in Power Grid Networks

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Submitted: April 12, 2026

Abstract

Still needs to be completed

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1 Introduction

1.1 Motivation

Current power transmission networks are complex, tightly coupled interconnected systems that must maintain stable frequency and voltage at all times, even under load fluctuations and communication delays or dropouts. Continuous feedback from every substation has become increasingly impractical as networks grow in size and complexity, given the rising telemetry, bandwidth, and overhead costs. This supports the concept of sparse, event-triggered coordination that sustains grid-wide synchronization with minimal actuation. In these systems, control impulses are applied intermittently, activated only when a measured error surpasses a specified threshold. The thresholds are established to ensure control while minimizing unnecessary updates and associated costs. The impulses are determined implicitly by the system’s evolution rather than being scheduled at fixed intervals. This introduces the challenge of preventing infinitely many triggers within a finite time, a phenomenon known as Zeno behavior [1]. This study uses event-triggered pinning impulse methods to guarantee synchronization under specified topological conditions, while deliberately preventing the occurrence of accumulation events.

1.2 Problem Introduction

Our study focuses on an event-triggered pinning impulsive control system. Instead of actuating each and every node, we actuate a small, preselected set of ‘pinned’ nodes; network coupling then propagates corrections to the rest of the grid. Each pinned node monitors a local error against a trigger threshold; when violated, a discrete impulse is applied, controlling the grid.

The proposed approach is informed by two pieces of prior research. First, the recent work on event-triggered pinning establishes sufficient conditions for global synchronization in complex dynamic networks [1]. It demonstrates how to select pinning nodes so that impulses at an arbitrary few locations control the overall dynamics, and proves convergence using Lyapunov analysis. Importantly, the event triggers rely solely on local error states, simplifying the amount of communication required for leader-follower dynamics while maintaining stability guarantees. Secondly, research on power-system synchronization constructed via Kuramoto-type oscillator networks establishes that the minimum coupling required for stabilization is a function of the graph structure [2]. It shows that weakly connected cuts serve as bottlenecks for connectivity and that small amounts of line removals will fundamentally change the stability threshold. Collectively, both studies support a topologically aware design structure and motivate pin selection in close proximity to the bottleneck.

More generally, within control theory literature, pinning control is proposed as a general strategy to control complex networks, while only requiring access to a subset of the nodes rather than every node in the system. While the type of complex network often varies, the two core questions are always: i) How many controllers are needed to ensure the network is controllable, and ii) Where to pin these controllers in order to achieve control [3]. These were core driving questions of the project.

Much of the literature within the space of pinning control often focuses on optimizing the structure/topology of a complex network to be suited for pinning control. This can be structured into four distinct optimization problems: optimizing the network with respect to the locations of the controllable nodes, optimizing the feedback gains, optimizing the two of these simultaneously, or optimizing the connection weights between nodes [4]. While these present interesting questions in the context of power grid design, when dealing with established infrastructure, the more pressing question becomes, for a given network topology, what is the most effective pinning strategy. As mentioned, a sufficient condition for the pinning strategy to ensure synchronization is outlined in [1], however, there is an outstanding question of, for power grids specifically, whether or not there exists a more effective and/or efficient pinning strategy. In the setting of power grids, this gap in pinning strategy literature becomes particularly appealing. Transmission networks are geographically sparse and either very costly or simply unfeasible to control at every node. As a result, control strategies that limit the necessary intervention have immense potential.

The other component of the proposed solution is the incorporation of event-triggered control. Similarly, this strategy has gained significant attention when dealing with power grids, particularly in cases where continuous communication between sparsely distribution nodes is impractical. Examples of this work include applying event-triggered load frequency control to lighten bandwidth requirements [5] and distributed

event-triggered control to resolve power sharing problems and reduce communication costs in microgrids [6]. Collectively, this work suggests that impulsive-control is well suited to power grid applications, when trying to achieve synchronization while under communication constraints.

Despite this well established literature base, much of the event-triggered control research applied to power systems focuses on frequency synchronization under pure event-triggered control. This leaves a meaningful gap when this theory is combined with the outlined pinning control strategy, summarizing the underlying question of the work: how can the topology based strategies of pinning control be combined with the communication-saving advantages and concepts of event-triggered control in order to achieve synchronization of power grid networks.

1.2.1 Kuramoto Model of Power Grids

We can model a power grid using N Kuramoto oscillators through the second-order Kuramoto equation given in [2]. Here, the phase $\theta_i(t) \in \mathbb{R}$ at node i is modelled by the following equation:

$$\alpha_i \ddot{\theta}_i + \dot{\theta}_i = \omega_i + \sum_{j=1}^N w_{ij} \sin(\theta_j - \theta_i) \quad (1)$$

where

- α_i is the inertia coefficient of node i
- ω_i is the natural frequency at node i
- w_{ij} is the coupling weight of edge (i, j)

and the nonlinear coupling term $\sum_j w_{ij} \sin(\theta_j - \theta_i)$ represents the synchronizing torque from each connected node acting on node i .

Synchronization in the Kuramoto model occurs when each node's instantaneous frequency $\dot{\theta}_i(t)$ converges to a common mean frequency Ω , defined as

$$\Omega = \frac{1}{N} \sum_{i=1}^N \omega_i. \quad (2)$$

Formally, frequency synchronization is achieved when

$$\lim_{t \rightarrow \infty} |\dot{\theta}_i(t) - \Omega| = 0, \quad \forall i \in \{1, 2, \dots, N\}. \quad (3)$$

Complete synchronization, where all nodes' phases coincide, corresponds to the stronger condition

$$\lim_{t \rightarrow \infty} |\theta_i(t) - \theta_j(t)| = 0, \quad \forall i, j. \quad (4)$$

More generally, phase-locked synchronization is obtained when the phase differences converge to constant values, which is usually (but not always) zero. In this state, all node frequencies satisfy $\dot{\theta}_i - \Omega \rightarrow 0$, ensuring coherent operation across the network. Frequency synchronization is critical: it enhances generation efficiency, maintains power quality, and supports overall grid stability. A lack of synchronization can damage electrical loads or, in severe cases, lead to cascading failures and grid-wide power outages.

This project aims to develop and validate an Event-Triggered Pinning Impulsive Control strategy to maintain frequency synchronization in modern power grids, with the objective of stabilizing the grid while minimizing operational costs and outage risks.

2 Theoretical Stability Analysis

2.1 Assumptions and Simplifications

In order to adequately analyze the presented synchronization problem in the given time frame from a theoretical standpoint, several assumptions and simplifications had to be made. Beginning with the second-order

Kuramoto model presented in (1), the first assumption is that the system operates near the synchronized state, which allows the dynamics to be expressed in a rotating reference frame and linearized about the equilibrium. From this, we obtain a linear state-space equation that we can conduct stability analysis on. In alignment with the previous literature which introduced the problem, it was also assumed that impulses conducted on the system act as idealized instantaneous jumps, a simplification which would not hold in real-world implementation. Additionally, the theoretical analysis introduces a global coupling strength constant c to scale the network Laplacian during continuous flow. It is important to emphasize that this does not imply uniform coupling across all nodes, as the intrinsic coupling strength is still represented within the Laplacian. While more advanced literature introduces dynamic/adaptive coupling laws between oscillators in the network ([1]), this simplified system allows for stability analysis of the global dynamics at an undergraduate level. As such, it is important to acknowledge that the theoretical results presented later represent stability findings for a simplified and idealized "power-grid-like" network, but are not reflective of the nonlinear power grid dynamics from which the problem stems.

2.2 Linearization of Kuramoto Equation

2.2.1 Model Introduction

Recall the second-order Kuramoto equation introduced in (1), where the phase $\theta_i(t) \in \mathbb{R}$ of node i is modelled by:

$$\alpha_i \ddot{\theta}_i + \dot{\theta}_i = \omega_i + \sum_{j=1}^N w_{ij} \sin(\theta_j - \theta_i) \quad (5)$$

where:

- α_i is the inertia coefficient of node i
- ω_i is the natural frequency at node i
- w_{ij} is the coupling weight of edge (i,j)

2.2.2 Linearization in Rotating Reference Frame

We first seek to linearize the model around the synchronized state, allowing for the use of traditional linear ODE tools to analyze the system. In order to do so, we recognize that when synchronized, the phase at each node θ_i is not zero, but rather, all phases rotate at some standard frequency of the system, Ω . With this, we define a rotating frame given by

$$\varphi_i(t) = \theta_i(t) - \Omega t. \quad (6)$$

Then, when in synchronization, we know that all phases can be expressed as

$$\theta_i^*(t) = \Omega t + \varphi_i^* \quad (7)$$

where φ_i^* is a constant. Subbing this into our rotating frame, we get that synchronization is equivalent to this fixed point

$$\varphi_i(t) = \theta_i^*(t) - \Omega t = \varphi_i^*. \quad (8)$$

Plugging the derivatives of (6) into (5), we end up with the same Kuramoto equation given in [2]:

$$\alpha_i \ddot{\varphi}_i + \dot{\varphi}_i = \omega_i + \sum_{j=1}^N w_{ij} \sin(\varphi_j - \varphi_i). \quad (9)$$

where the synchronized state is given by φ^* . In this synchronized state, it follows that $\dot{\varphi}_i^* = \ddot{\varphi}_i^* = 0$, and thus plugging into (9), when synchronized, we have

$$0 = \omega_i + \sum_{j=1}^N w_{ij} \sin(\varphi_j^* - \varphi_i^*). \quad (10)$$

Now, we can proceed with our linearization by allowing for a small deviation from our synchronized state at node i , given by $\delta_i(t)$. Then,

$$\varphi_i(t) = \varphi_i^* + \delta_i(t), \quad \dot{\varphi}_i(t) = \dot{\delta}_i(t), \quad \ddot{\varphi}_i(t) = \ddot{\delta}_i(t) \quad (11)$$

and plugging these into (9), we have

$$\alpha_i \ddot{\delta}_i + \dot{\delta}_i = \omega_i + \sum_{j=1}^N w_{ij} \sin((\varphi_j^* - \varphi_i^*) + (\delta_j - \delta_i)). \quad (12)$$

and by first-order Taylor expansion, this becomes

$$\alpha_i \ddot{\delta}_i + \dot{\delta}_i = \omega_i + \sum_{j=1}^N w_{ij} (\sin(\varphi_j^* - \varphi_i^*) + (\delta_j - \delta_i) \cos(\varphi_j^* - \varphi_i^*)). \quad (13)$$

Recalling our synchronized state condition in (10), this simplifies to

$$\alpha_i \ddot{\delta}_i + \dot{\delta}_i = \sum_{j=1}^N w_{ij} (\delta_j - \delta_i) \cos(\varphi_j^* - \varphi_i^*), \quad (14)$$

leaving us with our linearized system. Now, to better analyze the stability of our linearized system, we can first express the system in matrix form. Redefining our coupling strength as

$$w_{ij}^* = w_{ij} \cos(\varphi_j^* - \varphi_i^*), \quad (15)$$

which is constant, our linearized system can be written as

$$\alpha_i \ddot{\delta}_i + \dot{\delta}_i = \sum_{j=1}^N w_{ij}^* (\delta_j - \delta_i) = \sum_{j=1}^N w_{ij}^* \delta_j - \delta_i \sum_{j=1}^N w_{ij}^* \quad (16)$$

With this, we can introduce a Laplacian matrix, defined as

$$L_{ij} = \begin{cases} -w_{ij}^* & i \neq j \\ \sum_{k=1}^N w_{ik}^* & i = j \end{cases}$$

Then, we can rewrite (16) as

$$\alpha_i \ddot{\delta}_i + \dot{\delta}_i = - \sum_{j=1}^N L_{ij} \delta_j, \quad (17)$$

or expressing in terms of global vectors, letting $\delta = (\delta_1, \dots, \delta_N)^\top$ and $A = \text{diag}(\alpha_1, \dots, \alpha_N)$, we have

$$A \ddot{\delta} + \dot{\delta} + L \delta = 0. \quad (18)$$

Lastly, introducing a global scalar coupling gain constant c on the natural coupling term, we get our final linearized second-order system,

$$A \ddot{\delta} + \dot{\delta} + cL \delta = 0. \quad (19)$$

2.2.3 Linearized System in First-Order State Space Form

Given the linearized system derived above in (19), we can define global vectors as

$$\delta = \begin{bmatrix} \delta_1 \\ \vdots \\ \delta_N \end{bmatrix}, \quad \dot{\delta} = \begin{bmatrix} \dot{\delta}_1 \\ \vdots \\ \dot{\delta}_N \end{bmatrix}$$

and corresponding global state

$$X = \begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix} \in \mathbb{R}^{2N}.$$

Then, from (19), it follows that

$$\ddot{\delta} = -A^{-1}\dot{\delta} - cA^{-1}L\delta,$$

so expressing in state space form, we have

$$\dot{X} = \begin{bmatrix} \dot{\delta} \\ -cA^{-1}L\delta - A^{-1}\dot{\delta} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & I \\ -cA^{-1}L & -A^{-1} \end{bmatrix}}_{J(c)} X.$$

Thus, our system is fully described by

$$\dot{X} = J(c)X, \quad \text{where} \quad J(c) = \begin{bmatrix} 0 & I \\ -cA^{-1}L & -A^{-1} \end{bmatrix}. \quad (20)$$

2.3 Impulsive Pinning Law

We apply impulses to the pinned nodes at times $t_1 < t_2 < \dots$. At each impulse time t_k , the local state of each pinned node, $\mathcal{P}$, is scaled by a factor $1 - d_{imp} \in (0, 1)$:

$$\begin{bmatrix} \delta_i(t_k^+) \\ \dot{\delta}_i(t_k^+) \end{bmatrix} = (1 - d_{imp}) \begin{bmatrix} \delta_i(t_k^-) \\ \dot{\delta}_i(t_k^-) \end{bmatrix}, \quad i \in \mathcal{P}. \quad (21)$$

We can write this in state space form by defining a the following global matrix

$$G = \text{diag}(g_1, \dots, g_{2N}), \quad g_i = \begin{cases} 1 - d_{imp}, & i \in \mathcal{P} \cup (\mathcal{P} + N) \\ 1, & \text{otherwise} \end{cases} \quad (22)$$

where $\mathcal{P} + N := \{i + N : i \in \mathcal{P}\}$ accounts for the frequency component of the pinned nodes. Then, the impulse law can be simply expressed as

$$X(t_k^+) = G X(t_k^-). \quad (23)$$

To summarize, this leaves us with our complete hybrid closed-loop system, given by:

- **Continuous flows:** $\dot{X}(t) = J(c) X(t)$
- **Discrete jump:** $X(t_k^+) = G X(t_k^-)$

2.4 Stability Analysis

The following section covers the stability analysis of the linearized system introduced above. This is done in a couple key steps. First, a Lyapunov function is introduced and proven to be in fact valid. Then, LaSalle's Invariance Principle is applied to prove that the continuous system (no impulses) is stable. Next, the impulse law is reintroduced, and its impact on the Lyapunov function is examined. Lastly, by introducing a restriction on the impulse triggering condition, the final result is derived: stability of the hybrid system.

2.4.1 Lyapunov Analysis of Continuous System

Ignoring impulses, recall our continuous linearized system, given by

$$A\ddot{\delta} + \dot{\delta} + cL\delta = 0. \quad (24)$$

Note, synchronization corresponds to $\delta \in \text{span}\{\mathbf{1}\}$ (all nodes share the same phase shift, so there is no relative error), so the stability analysis must be conducted on the subspace

$$\mathcal{E} := \{\delta \in \mathbb{R}^N : \mathbf{1}^\top \delta = 0\}.$$

With this, we can introduce a Lyapunov function, stemming from its common form, and verify that it is indeed a valid Lyapunov function in the following lemma.

Lemma 2.1. *For any fixed coupling $c > 0$, the function*

$$V(\delta, \dot{\delta}) := \frac{1}{2} \dot{\delta}^\top A \dot{\delta} + \frac{c}{2} \delta^\top L \delta \quad (25)$$

is a valid Lyapunov function on the error subspace $\mathcal{E} \times \mathcal{E}$ for the system given in (24). Specifically:

1. *V is positive definite and radially unbounded on $\mathcal{E} \times \mathcal{E}$;*
2. *along any flow trajectory, $\dot{V}(\delta(t), \dot{\delta}(t)) = -\|\dot{\delta}(t)\|^2 \leq 0$.*

Proof. Since $A = \text{diag}(\alpha_i)$ with $\alpha_i > 0$, we have $A \succ 0$. Letting $\underline{\alpha} := \min \alpha_i > 0$, we get

$$\dot{\delta}^\top A \dot{\delta} \geq \underline{\alpha} \|\dot{\delta}\|^2.$$

Similarly, on the error subspace, we know L is positive definite. With this, letting $\underline{\lambda} := \lambda_{\min}(L)$, we have similarly that

$$\delta^\top L \delta \geq \underline{\lambda} \|\delta\|^2$$

where $\underline{\lambda} > 0$. So, substituting into (25), we get

$$V(\delta, \dot{\delta}) \geq \frac{1}{2} \underline{\alpha} \|\dot{\delta}\|^2 + \frac{c}{2} \underline{\lambda} \|\delta\|^2. \quad (26)$$

In particular, if $\delta \in \mathcal{E}$ and $V(\delta, \dot{\delta}) = 0$, then both $\delta = 0$ and $\dot{\delta} = 0$. Thus V is positive definite on $\mathcal{E} \times \mathcal{E}$. Additionally, if $\|(\delta, \dot{\delta})\| \rightarrow \infty$ with $\delta \in \mathcal{E}$, then $V(\delta, \dot{\delta}) \rightarrow \infty$, therefore V is radially unbounded.

To prove the second property, differentiating our function in (25), and noting that matrix L is symmetric,

$$\frac{d}{dt} \left(\frac{1}{2} \dot{\delta}^\top A \dot{\delta} \right) = \dot{\delta}^\top A \ddot{\delta}, \quad \frac{d}{dt} \left(\frac{c}{2} \delta^\top L \delta \right) = c \dot{\delta}^\top L \delta.$$

So, substituting the system definition $A\ddot{\delta} = -\dot{\delta} - cL\delta$, we get

$$\begin{aligned} \dot{V} &= \dot{\delta}^\top A \ddot{\delta} + c \dot{\delta}^\top L \delta \\ &= \dot{\delta}^\top (-\dot{\delta} - cL\delta) + c \dot{\delta}^\top L \delta \\ &= -\|\dot{\delta}\|^2 - c \dot{\delta}^\top L \delta + c \dot{\delta}^\top L \delta \end{aligned}$$

where the last two terms cancel, leaving

$$\dot{V} = -\|\dot{\delta}\|^2 \leq 0. \quad (27)$$

Therefore, we can conclude that V is a valid Lyapunov function. □

Now, in order to show stability, we can apply LaSalle's Invariance Principle, given in [7].

Theorem 2.2. (*LaSalle's Invariance Principle*): Let $\Omega \subset D$ be a compact set that is positively invariant with respect to the autonomous system $\dot{x} = f(x)$. Let $V : D \rightarrow \mathbb{R}$ be continuously differentiable and satisfy

$$\dot{V}(x) \leq 0 \quad \text{for all } x \in \Omega.$$

Define

$$E := \{x \in \Omega : \dot{V}(x) = 0\},$$

and let M denote the largest invariant set contained in E . Then every solution starting in Ω approaches M as $t \rightarrow \infty$.

Proof. See citation □

Lemma 2.3. *The continuous linear system*

$$\dot{X} = J(c)X, \quad X = \begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix} \in \mathcal{E} \times \mathcal{E}$$

satisfies all hypotheses of LaSalle's Invariance Principle (Theorem 3.2). Consequently, every solution starting in $\mathcal{E} \times \mathcal{E}$ converges to the origin $(0, 0)$ as $t \rightarrow \infty$.

Proof. In order to apply LaSalle's, we must verify our system satisfies these conditions. First, recall our continuous system is independent of time, thus the system is trivially autonomous.

Now, recall our Lyapunov function

$$V(\delta, \dot{\delta}) = \frac{1}{2} \dot{\delta}^\top A \dot{\delta} + \frac{c}{2} \delta^\top L \delta.$$

For a fixed initial condition $(\delta(0), \dot{\delta}(0))$, define the set

$$\Omega := \left\{ (\delta, \dot{\delta}) \in \mathcal{E} \times \mathcal{E} : V(\delta, \dot{\delta}) \leq V(\delta(0), \dot{\delta}(0)) \right\}.$$

Positive invariance: From verifying that our Lyapunov function is valid, we have from (27) that

$$\dot{V}(\delta, \dot{\delta}) = -\|\dot{\delta}\|^2 \leq 0$$

and so V is non-increasing. Therefore, if our initial state is in Ω , then

$$V(\delta(t), \dot{\delta}(t)) \leq V(\delta(0), \dot{\delta}(0)) \quad \text{for all } t \geq 0,$$

so it remains in Ω for all time and thus Ω is positively invariant.

Compactness: Since V is continuous and Ω is the set of all points where V is less than or equal to a constant, Ω is closed. Now we show Ω is bounded. From the inequality given in (26), for any $\delta \in \mathcal{E}$, we have that

$$V(\delta, \dot{\delta}) \geq \frac{1}{2} \underline{\alpha} \|\dot{\delta}\|^2 + \frac{c}{2} \underline{\lambda} \|\delta\|^2.$$

Now, if $(\delta, \dot{\delta}) \in \Omega$, by definition

$$V(\delta, \dot{\delta}) \leq V(\delta(0), \dot{\delta}(0)) =: V_0.$$

Combining these bounds, we have

$$\frac{1}{2} \underline{\alpha} \|\dot{\delta}\|^2 + \frac{c}{2} \underline{\lambda} \|\delta\|^2 \leq V_0$$

which implies that

$$\|\dot{\delta}\|^2 \leq \frac{2V_0}{\underline{\alpha}} \quad \|\delta\|^2 \leq \frac{2V_0}{c\underline{\lambda}}$$

Therefore both δ and $\dot{\delta}$ are bounded, therefore Ω is bounded, so finally, we can conclude that Ω is compact.

With this, we can now define

$$E := \{(\delta, \dot{\delta}) \in \Omega : \dot{V}(\delta, \dot{\delta}) = 0\}.$$

Recall again from (27), that we have

$$\dot{V} = -\|\dot{\delta}\|^2,$$

thus, it follows that

$$\dot{V} = 0 \iff \dot{\delta} = 0.$$

So, the set E simplifies to

$$E = \{(\delta, \dot{\delta}) \in \Omega : \dot{\delta} = 0\}.$$

Now, if we assume that a solution remains in E for all time, it implies that $\dot{\delta}(t) = 0$, and therefore $\ddot{\delta}(t) = 0$ for all t . Substituting these into (24), this gives

$$cL\delta(t) = 0.$$

Since $c > 0$ and $L \succ 0$, this implies that $\delta(t) = 0$. Hence the largest invariant set contained in E is

$$M = \{(0, 0)\}.$$

Since we have shown that all conditions of LaSalle's Invariance Principle are satisfied, we can conclude that any solution starting in $\mathcal{E} \times \mathcal{E}$ converges to $(0, 0)$ as $t \rightarrow \infty$. Since the system is linear time-invariant on this subspace, asymptotic stability implies exponential stability, or equivalently, that $J(c)$ is Hurwitz on the error subspace. $\square$

Theorem 2.4. *The matrix $J(c)$ describing the linearized system*

$$X = \begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix} \in \mathcal{E} \times \mathcal{E}, \quad \dot{X} = J(c)X, \quad J(c) = \begin{bmatrix} 0 & I \\ -cA^{-1}L & -A^{-1} \end{bmatrix}$$

is Hurwitz on the error subspace.

Proof. By Lemma 2.3, every solution $X(t) = (\delta(t), \dot{\delta}(t))$ of the system $\dot{X} = J(c)X$ with initial condition in $\mathcal{E} \times \mathcal{E}$ satisfies

$$X(t) \rightarrow (0, 0) \quad \text{as } t \rightarrow \infty.$$

Consider an arbitrary eigenvalue $\lambda \in \mathbb{C}$ of $J(c)$ with corresponding eigenvector $v \neq 0$, $v \in \mathbb{C}^{2N}$. The function

$$X(t) = e^{\lambda t}v$$

is a solution of $\dot{X} = J(c)X$. Taking the norm,

$$\|X(t)\| = |e^{\lambda t}| \cdot \|v\| = e^{(\operatorname{Re} \lambda)t} \|v\|.$$

Suppose, for contradiction, that $\operatorname{Re} \lambda \geq 0$ for some eigenvalue λ . Then

$$\|X(t)\| = e^{(\operatorname{Re} \lambda)t} \|v\| \geq \|v\| > 0$$

for all $t \geq 0$, so $X(t) \not\rightarrow 0$ as $t \rightarrow \infty$. This contradicts the global asymptotic stability of the origin. Thus, $\operatorname{Re} \lambda < 0$ for every eigenvalue λ of J , hence $J(c)$ is Hurwitz. $\square$

2.4.2 Introduction of Impulses on Norm

Next, we introduce the impulses to the system. Conceptually, when a pinned node $i \in \mathcal{P}$ receives an impulse, we scale its local error state by a factor strictly between 0 and 1, while all other nodes are unchanged. To monitor the total error in the system, we can introduce

$$\|X\|^2 = \sum_{i=1}^N \|e_i\|^2, \quad e_i := \begin{bmatrix} \delta_i \\ \dot{\delta}_i \end{bmatrix} \in \mathbb{R}^2.$$

With this, we have our impulse law, which is that for a pinned node $i \in \mathcal{P}$,

$$e_i^+ = (1 - d_{\text{imp}}) e_i^-, \quad e_j^+ = e_j^- \quad (j \neq i), \quad d_{\text{imp}} \in (0, 1),$$

where the superscripts $-$ and $+$ denote the times immediately before and after the impulse, respectively.

Lemma 2.5. *Given our impulse law (above),*

$$\|X^+\|^2 \leq \|X^-\|^2,$$

with equality if and only if $e_i^- = 0$.

Proof. By definition,

$$\|X\|^2 = \sum_{j=1}^N \|e_j\|^2.$$

and for the pinned node i , we have $e_j^+ = e_j^-$ for all $j \neq i$, hence all terms in the sum cancel except the i th, so

$$\|X^+\|^2 - \|X^-\|^2 = \|e_i^+\|^2 - \|e_i^-\|^2.$$

Substituting $e_i^+ = (1 - d_{\text{imp}})e_i^-$

$$\|e_i^+\|^2 = \|(1 - d_{\text{imp}})e_i^-\|^2 = (1 - d_{\text{imp}})^2 \|e_i^-\|^2,$$

so

$$\|X^+\|^2 - \|X^-\|^2 = ((1 - d_{\text{imp}})^2 - 1) \|e_i^-\|^2.$$

Expanding the coefficient gives

$$(1 - d_{\text{imp}})^2 - 1 = 1 - 2d_{\text{imp}} + d_{\text{imp}}^2 - 1 = -d_{\text{imp}}(2 - d_{\text{imp}}).$$

Since $d_{\text{imp}} \in (0, 1)$, we have $d_{\text{imp}} > 0$ and $2 - d_{\text{imp}} > 1$, hence

$$-d_{\text{imp}}(2 - d_{\text{imp}}) < 0.$$

Therefore

$$\|X^+\|^2 - \|X^-\|^2 = -d_{\text{imp}}(2 - d_{\text{imp}}) \|e_i^-\|^2 \leq 0,$$

with equality if and only if $\|e_i^-\|^2 = 0$, that is $e_i^- = 0$. $\square$

2.4.3 Event Triggered Impulsive Pinning

With the effect of impulses described, we now introduce the event-triggering rule for impulsive control. As we allow the impulses to increase the Lyapunov function, we must ensure that impulses do not de-stabilize the system over time. With that in mind, let $P \succ 0$ be the unique solution to the Lyapunov equation

$$J(c)^\top P + PJ(c) = -I,$$

and define $\eta := \lambda_{\max}(P)/\lambda_{\min}(P) \geq 1$. Then the impulse times $t_1 < t_2 < \dots$ are determined by the *event-triggering rule* such that an impulse occurs at t_{k+1} only when the state error has decayed to satisfy

$$V(X(t_{k+1}^-)) = \mu V(X(t_k^+)), \quad (28)$$

for a chosen design parameter $\mu \in (0, 1/\eta)$. We may visualize the proposed behavior,

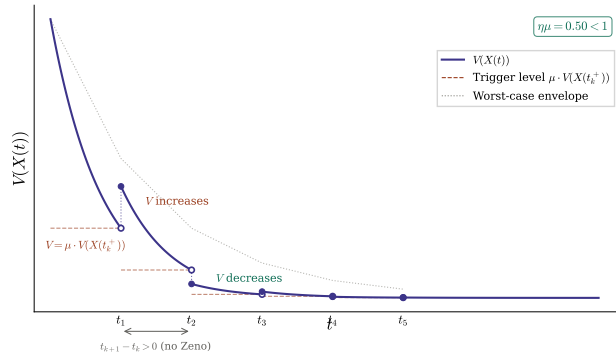


Figure 1: Illustrative event-triggered impulse trajectory showing mixed jump behavior.

Lemma 2.6. *Consider the hybrid system with constant coupling $c > 0$, given by*

$$\dot{X} = J(c)X,$$

with the event-triggered impulsive pinning described above. Then, Zeno behavior is strictly excluded and there exists τ_{max} such that $t_{k+1} - t_k \leq \tau_{max}$.

Proof. Fix $c > 0$. From Lemma 2.3, $J(c)$ is Hurwitz on the error subspace. Thus, there exists a unique $P \succ 0$ solving $J(c)^\top P + PJ(c) = -I$. Defining $V(X) := X^\top PX$, the derivative along the continuous flow is

$$\dot{V}(X) = -\|X\|^2 \leq 0.$$

Using the Rayleigh quotient bounds $\lambda_{\min}(P)\|X\|^2 \leq V(X) \leq \lambda_{\max}(P)\|X\|^2$, we can bound the continuous decay rate from both above and below. Letting $\alpha := \frac{1}{\lambda_{\max}(P)} > 0$ and $\Gamma := \frac{1}{\lambda_{\min}(P)} > 0$, we have

$$-\Gamma V(X) \leq \dot{V}(X) \leq -\alpha V(X). \quad (29)$$

Integrating these bounds over any continuous flow interval $t \in [t_k, t_{k+1})$ yields

$$V(X(t_k^+))e^{-\Gamma(t-t_k)} \leq V(X(t)) \leq V(X(t_k^+))e^{-\alpha(t-t_k)}. \quad (30)$$

At the trigger time t_{k+1} , condition (28) gives $V(X(t_{k+1}^-)) = \mu V(X(t_k^+))$. Substituting this into the lower bound of (30) gives

$$\mu V(X(t_k^+)) \geq V(X(t_{k+1}^-))e^{-\Gamma(t_{k+1}-t_k)}.$$

Assuming $X \neq 0$ (otherwise the system is trivially at the origin and no impulses occur), we divide by $V(X(t_k^+))$ and take the natural logarithm to find a strict lower bound on the inter-event time

$$t_{k+1} - t_k \geq \frac{\ln(1/\mu)}{\Gamma} =: \tau_{\min} > 0. \quad (31)$$

Since the inter-event time is bounded below by a strictly positive constant $\tau_{\min}$, Zeno behavior is strictly excluded.

Similarly, substituting the trigger condition into the upper bound of (30) yields

$$\mu V(X(t_k^+)) \leq V(X(t_{k+1}^-))e^{-\alpha(t_{k+1}-t_k)} \implies t_{k+1} - t_k \leq \frac{\ln(1/\mu)}{\alpha} =: \tau_{\max}.$$

Thus, the system is guaranteed to reach the event trigger in finite time. $\square$

2.4.4 Stability of Hybrid System

Theorem 2.7. *Consider the hybrid system with constant coupling $c > 0$, given by*

$$\dot{X} = J(c)X$$

with event-triggered impulsive pinning on the set $\mathcal{P}$. Let $P \succ 0$ be the unique solution to the Lyapunov equation $J(c)^\top P + PJ(c) = -I$ and define $\eta := \lambda_{\max}(P)/\lambda_{\min}(P) \geq 1$.

Assume the impulse times $t_1 < t_2 < \dots$ are determined by an event-triggering rule such that an impulse occurs at time t_{k+1} only when the state error has decayed to satisfy

$$V(X(t_{k+1}^-)) = \mu V(X(t_k^+)), \quad (32)$$

for a chosen parameter $\mu \in (0, 1/\eta)$. Then, Zeno behavior is strictly excluded, and the system synchronization is globally exponentially stable on the error subspace.

Proof. Across any jump time t_k , Lemma 2.5 establishes that $\|X^+\|^2 \leq \|X^-\|^2$. We may relate this to the Lyapunov function $V = X^\top P X$ using the Rayleigh quotient bounds used in Lemma 3.6

$$\lambda_{\min}(P)\|X\|^2 \leq V(X) \leq \lambda_{\max}(P)\|X\|^2. \quad (33)$$

Specifically,

$$\begin{aligned} V(X(t_k^+)) &\leq \lambda_{\max}\|X(t_k^+)\|^2 && \text{(Rayleigh upper bound)} \\ &\leq \lambda_{\max}\|X(t_k^-)\|^2 && \text{(Lemma 2.5)} \\ &\leq \frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}V(X(t_k^-)) && \text{(Rayleigh lower bound)} \\ &= \eta V(X(t_k^-)) && \text{(Using } \eta = \lambda_{\max}(P)/\lambda_{\min}(P)) \end{aligned}$$

Substituting in the event trigger condition (32), the Lyapunov function evaluated immediately after the $(k+1)$ -th jump satisfies

$$V(X(t_{k+1}^+)) \leq \eta V(X(t_{k+1}^-)) \leq \eta\mu V(X(t_k^+)).$$

Let $\rho := \eta\mu$; then as $\mu \in (0, 1/\eta)$, it follows that $\rho \in (0, 1)$. Then, with the initial condition $t_0 = 0$, we may inductively conclude the post-jump sequence decays geometrically

$$V(X(t_k^+)) \leq \rho V(X(t_{k-1}^+)) \leq \dots \leq \rho^k V(X(0)). \quad (34)$$

In order to translate this jump-dependent bound into a time-dependent bound, recall the maximum inter-event time $\tau_{\max} := \frac{\ln(1/\mu)}{\alpha}$, proved in Lemma 2.6. Using this, since $t \leq (k+1)\tau_{\max}$, we have $k \geq \frac{t}{\tau_{\max}} - 1$, and because $\rho \leq 1$, we can bound ρ^k above by

$$\rho^k \leq \rho^{\frac{t}{\tau_{\max}} - 1} = \frac{1}{\rho} e^{-\beta t}, \quad \text{where } \beta := \frac{\ln(1/\rho)}{\tau_{\max}} > 0.$$

Hence, the Lyapunov function is a decaying exponential

$$V(X(t)) \leq \frac{1}{\rho} e^{-\beta t} V(X(0)).$$

We may now chain inequalities to translate the decay of the Lyapunov function back into state norms,

$$\begin{aligned} \|X(t)\|^2 &\leq \frac{1}{\lambda_{\min}(P)} V(X(t)) && \text{(Rayleigh lower bound)} \\ &\leq \frac{1}{\lambda_{\min}(P)} \left(\frac{1}{\rho} e^{-\beta t} V(X(0)) \right) && \text{(Exponential decay of } V) \\ &\leq \frac{1}{\lambda_{\min}(P)} \frac{1}{\rho} e^{-\beta t} (\lambda_{\max}(P)\|X(0)\|^2) && \text{(Rayleigh upper bound at } t=0) \\ &= \frac{\eta}{\rho} e^{-\beta t} \|X(0)\|^2 && \text{(Using } \eta = \lambda_{\max}(P)/\lambda_{\min}(P)) \\ &= \frac{1}{\mu} e^{-\beta t} \|X(0)\|^2 && \text{(Using } \rho = \eta\mu) \end{aligned}$$

Thus, we arrive at the final bound at the state norm

$$\|X(t)\|^2 \leq \frac{1}{\mu} e^{-\beta t} \|X(0)\|^2.$$

Taking the square root and defining $C := 1/\sqrt{\mu}$ we are left with the following result

$$\|X(t)\| \leq C e^{-(\beta/2)t} \|X(0)\|.$$

This proves that the origin $(0,0)$ on the error subspace $\mathcal{E} \times \mathcal{E}$ is globally exponentially stable under the proposed event-triggering rule. $\square$

2.5 Numerical Verification

To test the theoretical results, we use a network modelled after the IEEE 14-bus test system **(cite)**. The network consists of $N = 14$ nodes connected by 20 transmission lines. The topology and coupling weights are shown in Figure 2. The inertia coefficients are taken to be uniform at $\alpha_i = 0.5$ for all i , and the global coupling gain is set to $c = 0.3$.

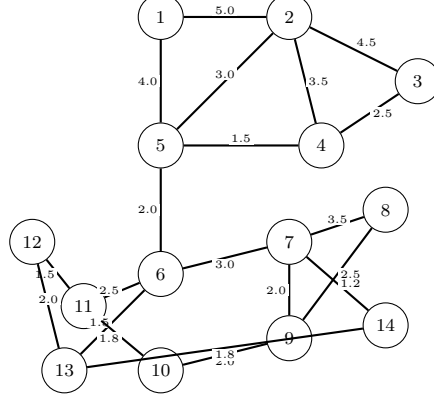


Figure 2: Network topology based on the IEEE 14-bus test system. Edge labels denote the linearized coupling weights w_{ij}^* .

2.5.1 Lyapunov Analysis and Event-Triggering Design

Solving the Lyapunov equation $J(c)^\top P + PJ(c) = -I$ numerically yields $P \succ 0$ with

$$\lambda_{\min}(P) \approx 0.154, \quad \lambda_{\max}(P) \approx 5.296, \quad \eta = \frac{\lambda_{\max}(P)}{\lambda_{\min}(P)} \approx 20.1.$$

Hence, the event-triggering parameter must satisfy $\mu < 1/\eta \approx 0.0498$. We consider three design choices:

- **Aggressive:** $\mu = 0.0494$, giving $\eta\mu = 0.99$ (near the stability boundary);
- **Moderate:** $\mu = 0.0249$, giving $\eta\mu = 0.50$;
- **Conservative:** $\mu = 0.0124$, giving $\eta\mu = 0.25$.

All 14 nodes are pinned with impulse gain $d_{\text{imp}} = 0.9$.

2.5.2 Simulation Results

The hybrid system is simulated from a random initial value $X(0) \in \mathcal{E} \times \mathcal{E}$ with $\|X(0)\| \approx 1.37$. Figure 3 compares the Lyapunov function and state norm trajectories across the three triggering strategies and the continuous baseline.

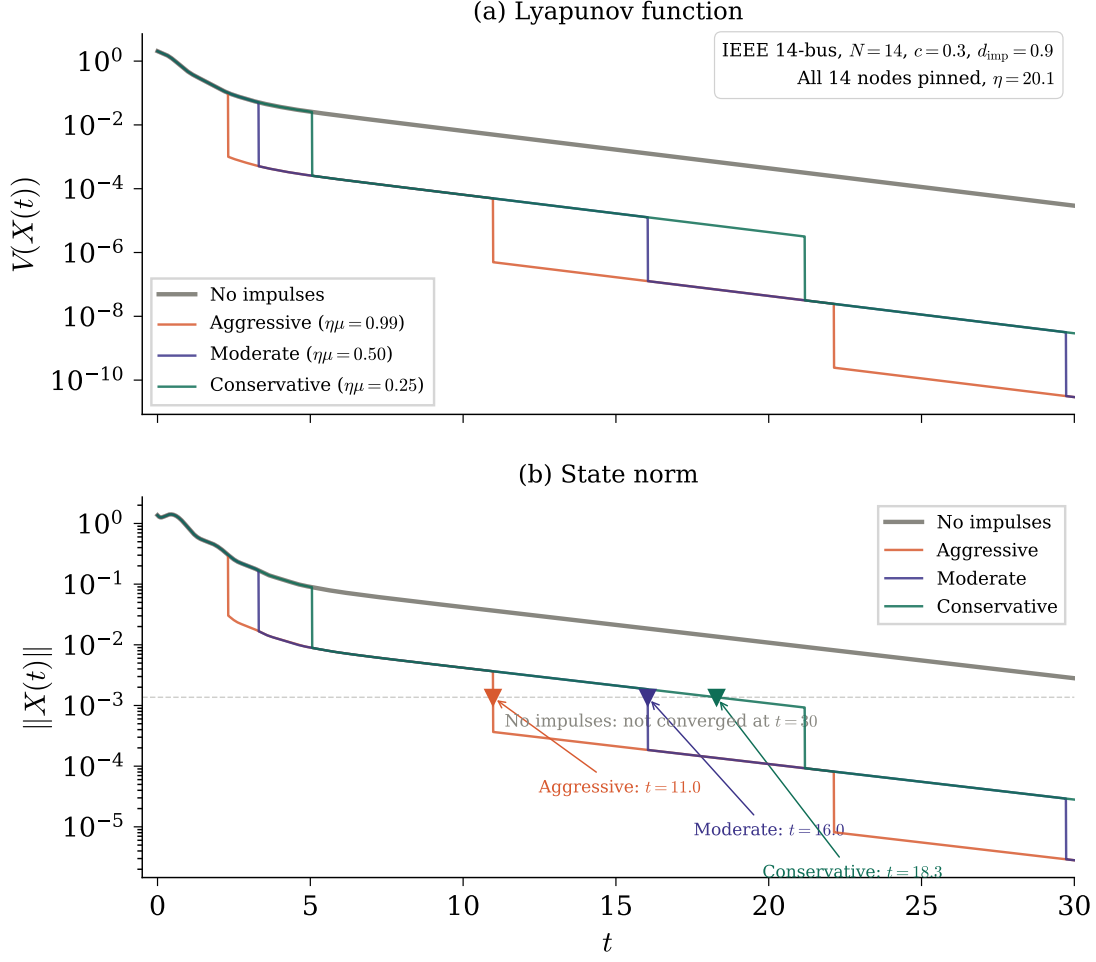


Figure 3: Convergence comparison on the IEEE 14-bus network. (a) Lyapunov function $V(X(t))$ on a log scale. (b) State norm $\|X(t)\|$ with the convergence threshold $10^{-3}\|X(0)\|$ shown as a dashed line.

The results are summarized in Table 1.

Strategy	$\eta\mu$	Impulses	Conv. time
No impulses	—	—	> 30
Aggressive	0.99	3	11.0
Moderate	0.50	3	16.0
Conservative	0.25	2	18.3

Table 1: Convergence comparison ($\|X(t)\| < 10^{-3}\|X(0)\|$) for the IEEE 14-bus network.

3 Simulation Background

3.1 Simulation Mathematical Background

3.1.1 Linearized Dynamics

From Section 3, the linearized system with the Kuramoto dynamics can be written as:

$$\alpha_i \ddot{\delta}_i + \dot{\delta}_i = \sum_{j=1}^N w_{ij}^* (\delta_j - \delta_i) = \sum_{j=1}^N w_{ij}^* \delta_j - \delta_i \sum_{j=1}^N w_{ij}^* \quad (35)$$

With the Laplacian matrix:

$$L_{ij} = \begin{cases} -w_{ij}^* & i \neq j \\ \sum_{k=1}^N w_{ik}^* & i = j \end{cases}$$

3.1.2 Simulation Update Step

The power grid simulations use the linearized Kuramoto dynamics to update the state vector X . The state vector tracks the phase and frequency deviations at the current time step:

$$\Delta\theta = \delta, \quad \Delta\omega = \dot{\delta}, \quad X = \begin{bmatrix} \Delta\theta \\ \Delta\omega \end{bmatrix} = \begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix}. \quad (36)$$

At each time step, calculate: $\dot{\Delta\theta}$ and $\dot{\Delta\omega}$:

$$\dot{\Delta\theta}_1 = \Delta\omega, \quad (37)$$

,

$$\dot{\Delta\omega} = -M^{-1}D(\Delta\omega) - c(t)M^{-1}L_w(\Delta\theta). \quad (38)$$

Which are used to update the state vector X with the update step:

$$X = X * \begin{bmatrix} \dot{\Delta\theta} \\ \dot{\Delta\omega} \end{bmatrix} \quad (39)$$

In the simulations, we used a simplified Laplacian matrix:

$$L_w \approx L_{approx}, \quad (40)$$

where L_{approx} is the Laplacian matrix formed directly from the line coupling a_{ij} :

$$w_{ij} = a_{ij} \cos(\varphi_i^* - \varphi_j^*) \approx a_{ij}, \quad (41)$$

which is an acceptable approximation when the synchronized phase differences are small.

$$\cos(\varphi_i^* - \varphi_j^*) \approx 1, \quad (42)$$

3.1.3 Centrality-based selection of pinned buses

Theorem 1 in [1] introduces the sufficient condition that guarantees the synchronization of a network. It states that synchronization can be achieved if:

$$\Lambda = \gamma I_{N-l} + c\bar{A} \prec 0,$$

where N is the number of nodes, l is the number of pinned nodes, $\gamma = \|K\|$ is the Lipschitz bound, and $\bar{A}$ is the reduced coupling matrix associated with the unpinned nodes. Since $\bar{A}$ is symmetric, this is equivalent to

$$\lambda_{\max}(\bar{A}) < -\frac{\gamma}{c},$$

or, equivalently,

$$c > \frac{\gamma}{-\lambda_{\max}(\bar{A})}.$$

Hence, our goal is to find pinned set P such that $\bar{A}$ is sufficiently negative in the spectral sense.

In the simulations we apply the following iterative algorithm to determine which nodes to pin:

1. Start with $P = \emptyset$

2. Evaluate $\lambda_{\max}(\bar{A})$
3. If $\lambda_{\max}(\bar{A}) < -\frac{\gamma}{c}$, the synchronization criterion is met and we can stop.
4. Otherwise, add the next free node with the lowest degree and repeat. In this step, degree is calculated as:

$$\text{degree}(i) = |\{j : a_{ij} \neq 0\}| \quad (43)$$

3.2 Simulation limitations

The simulation and applied control methods are built around the Kuramoto equation linearized around a stable operating point, meaning the current results are only valid for deviations where the small angle approximation $\sin \theta \approx \theta$ is within acceptable amounts of error. At $\theta = 30^\circ$, the error is $\sim 0.023^\circ$.

The current simulation treats all nodes in the network as controllable (and thus pinnable), which would not be the case on a real power grid. Residential areas, for example, cannot be controlled. The grid can be controlled from power stations, phase-shifting transformers, and High Voltage DC connections. Even within power sources, not all types of power sources are good fits for impulsive control. Synchronous generators have inertia that limits their ability to quickly change output level. The network coupling is also simplified, with the entries of the weighted Laplacian defined as $\frac{1}{x_{ij}}$

All nodes are treated as being capable of receiving the same magnitude of impulse, when real power sources have different limitations on how rapidly the angular velocity can be changed. Synchronous generators have inertia due to the rotor's mass that affects their ability to accelerate and decelerate.

3.3 Shared Simulation Skeleton

The two main simulations are not intended to test different dynamical models, but rather different network cases under the same previously derived framework. Both implementations inherit the linearized second-order Kuramoto state-space formulation introduced above, including the same stacked phase-frequency state definition, the same approximated weighted Laplacian, and the same explicit update structure for propagating the state forward in time. In other words, the governing equations are held fixed across both implementations so that later differences in convergence, control effort, or disturbance response can be interpreted primarily as consequences of topology and scenario design rather than of a change in the underlying model.

That common modeling choice also means that both simulations inherit the same simplifying assumptions already stated in the preceding derivation. In particular, both rely on the small-angle linearization and on the approximation of the weighted Laplacian by the simplified coupling matrix used in the implementation. Rather than repeating those limitations here, the important point for comparison is that they are shared equally by both case studies. This gives the later comparison a more controlled interpretation: neither simulation is granted a more detailed physical model than the other, so the comparison isolates structural differences in the network and control setup while accepting the same modeling simplifications in both cases.

The same logic applies to the simulation update step. Both implementations use the state update rule developed in the previous section: the current state is decomposed into phase and frequency deviations, the local synchronization error is evaluated relative to the chosen reference trajectory, any event-triggered impulsive correction is applied, and the continuous dynamics are then advanced to the next time step. This common ordering is important because it ensures that both simulations realize the same hybrid control architecture, namely continuous linearized dynamics between events and discrete state contractions at event times. As a result, the comparison between the two simulations is not a comparison between different numerical procedures, but between different network realizations of the same control procedure.

The pinning structure should also be interpreted as inherited from the earlier theory rather than redefined here. Theorem 1 provides the synchronization criterion by requiring that the reduced unpinning network satisfy the relevant spectral condition. The simulations then implement this criterion through a practical

iterative node-selection rule: beginning from the current pinned set, the next free node with the lowest degree is added until the theorem-based stopping condition is met. Framed this way, low-degree selection is not the theorem itself, but the algorithmic rule used to satisfy the theorem’s synchronization requirement within each network. This distinction is important because it separates the analytical guarantee from the implementation strategy.

Keeping that theorem-guided pinning logic fixed across both simulations is methodologically useful. It means that both case studies are built from the same synchronization criterion and the same low-degree-first node addition rule, even though the actual pinned sets differ because the underlying topologies differ. This allows later discussion to focus on how the theorem-guided strategy behaves under a structured power-grid network versus a synthetic random network, instead of being confounded by different node-selection philosophies. The same idea extends to topology loss or blackout scenarios: the pinned set may be recomputed when the network changes, but it is recomputed according to the same criterion and the same practical selection rule.

The event-triggered impulsive controller is likewise shared at the framework level. Both simulations use the same exponential threshold structure for deciding when a pinned node should receive an impulse, and both apply the same local contraction-type jump once the trigger condition is met. The purpose of keeping this mechanism fixed is to preserve a common measure of control aggressiveness across both implementations. In effect, the two simulations are being asked the same control question: under the same event-triggered impulsive law, how does synchronization behavior change when the network structure changes? This is a stronger comparison than simply observing two different codes, because the trigger logic and reset structure are held conceptually constant.

Finally, both implementations record the same class of outputs, including phase trajectories, frequency trajectories, aggregate error-energy measures, and impulse counts. This shared output structure is what makes the later side-by-side interpretation meaningful. Since both codes are built on the same derived model, the same update logic, the same theorem-guided pinning framework, and the same event-triggered impulse structure, the recorded differences can be read as differences in network response rather than differences in what was measured. For that reason, this subsection should be understood as the analytical bridge between the derivation above and the two implementation-specific subsections that follow: the mathematics establishes the common framework, while the next subsections show how that framework is instantiated in each network case.

4 Simulation Results

4.1 100 Node Sections

To verify that the linearized Kuramoto model agrees with the theoretical results, first a simulation was run on a 100 node network with random edges and edge weights. Figure 4 shows the topology. The nodes were pinned according to the centrality based pinning algorithm and convergence criteria set out in [1].

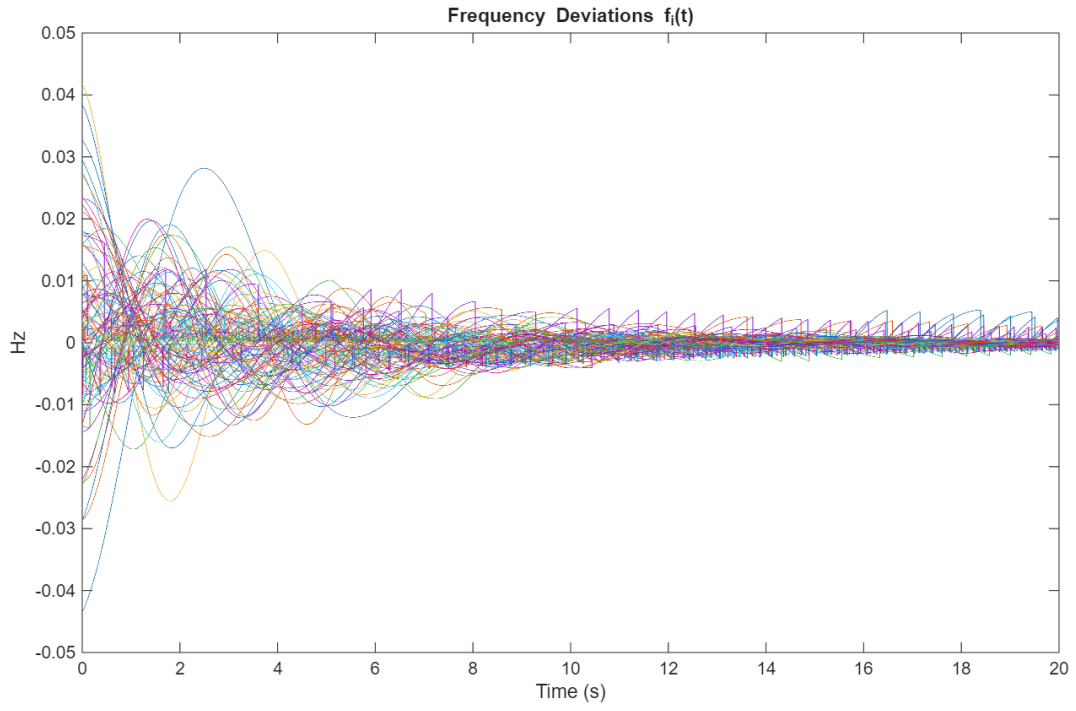


Figure 5

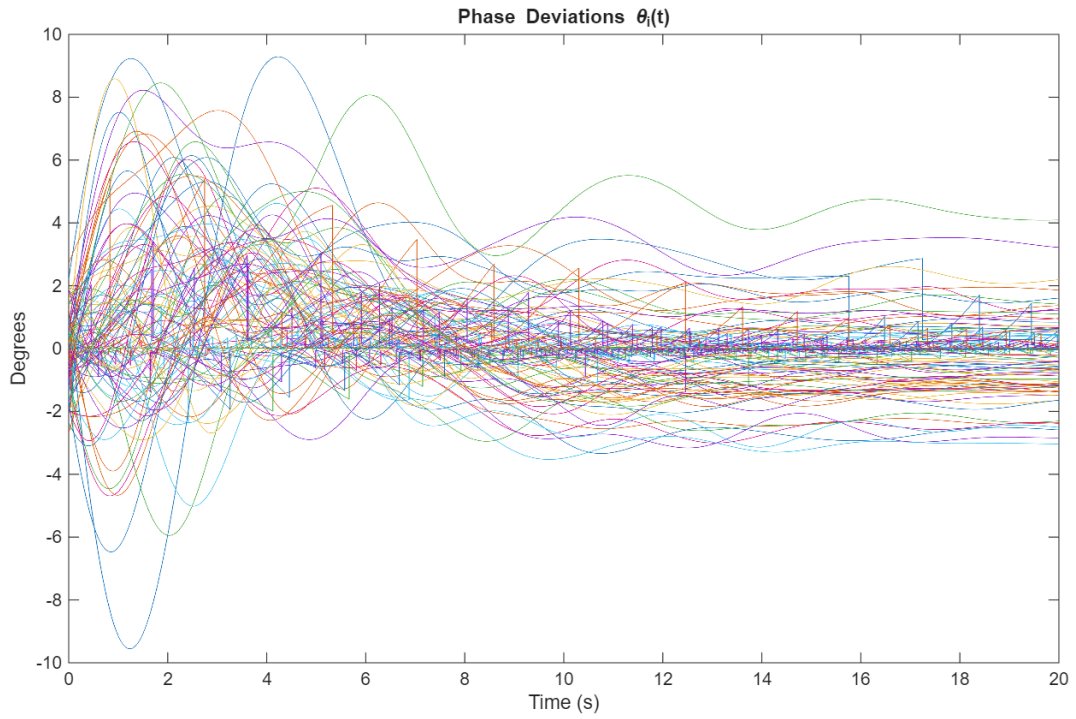


Figure 6

Figure 7 shows the combined lyapunov error of the angular velocities and phase angles converging. Since the phase angles converge to their respective equilibria positions, it converges to some $y > 0$.

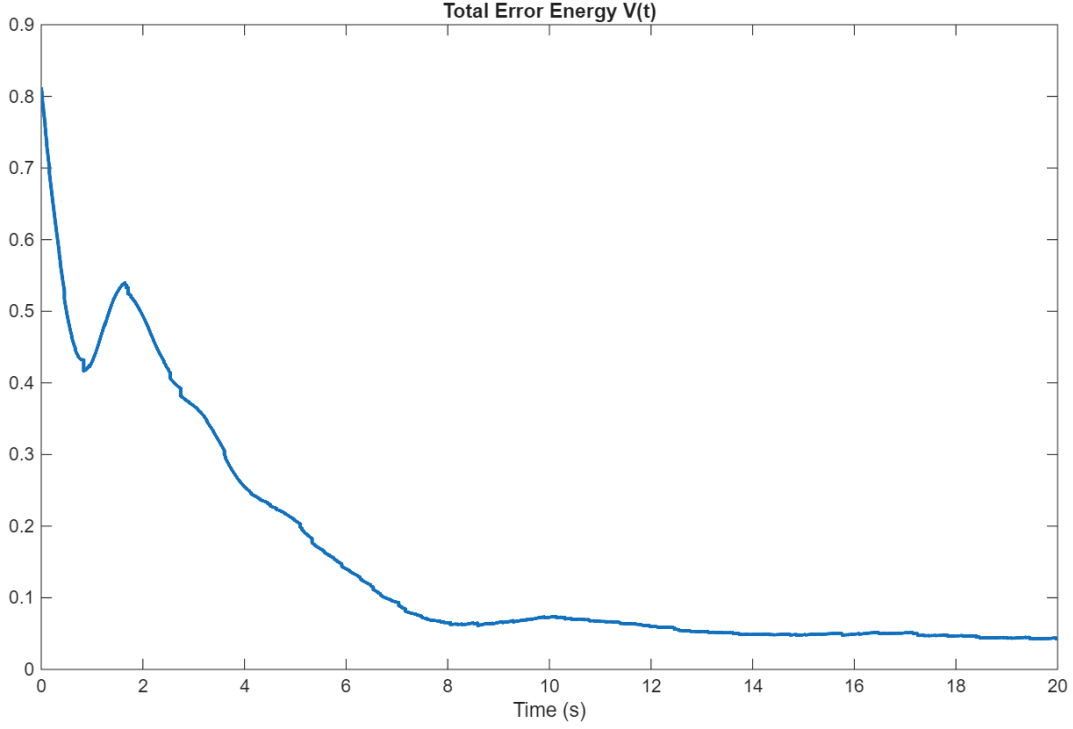


Figure 7: Lyapunov Error $V(t) = \sum_{i=1}^{100} e_i^T(t)e_i(t)$

Figure 8 shows most nodes receiving under 150 impulses, with node 47 receiving 500.

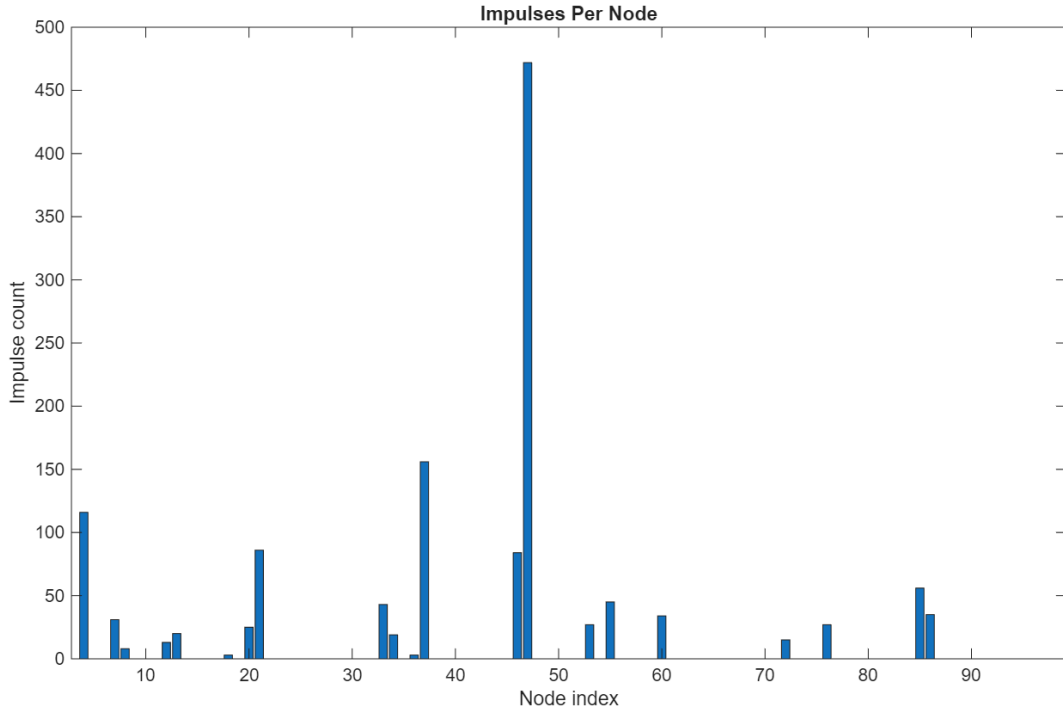


Figure 8: Total Impulses applied to each pinned node

These simulations agree with the theoretical convergence criteria. Of note, however is that applying impulses to the phase may be causing unnecessary impulses and delaying convergence, as the phases are not

meant to be corrected to 0, but instead to their unique offset at the stable operating point for the network. A simulation done only applying impulses to frequency can be seen in Section 7.0.1, motivating future work in studying different stability criteria.

4.2 IEEE 39-Bus System Simulations

So far we have discussed the results of event triggered pinning control in power grids with a random connected grid topologies. Additionally, the previous power grid topologies assumed uniform damping and inertia parameters. To further demonstrate the effectiveness of the control strategy on power grids, we analyzed the performance of the control strategy using a real power grid topology. The IEEE 39-bus system topology was used for this analysis.

4.2.1 System Data and Topology

IEEE 39-Bus System

The IEEE 39-bus system is a sample power grid model, commonly known as the 10-machine New England Power System. The model is popular among researchers working on transient stability analysis, speed controller analysis, and system frequency analysis [8]. This is because the system is large enough to exhibit complex power grid behavior, but small enough to maintain acceptable simulation times [9].

Topology

The model has 39 busses, where busses 1-29 are load busses, and busses 30-39 are generator busses. There are 46 transmission lines (connections between nodes in the topology) in the network. See **Figure 9**.

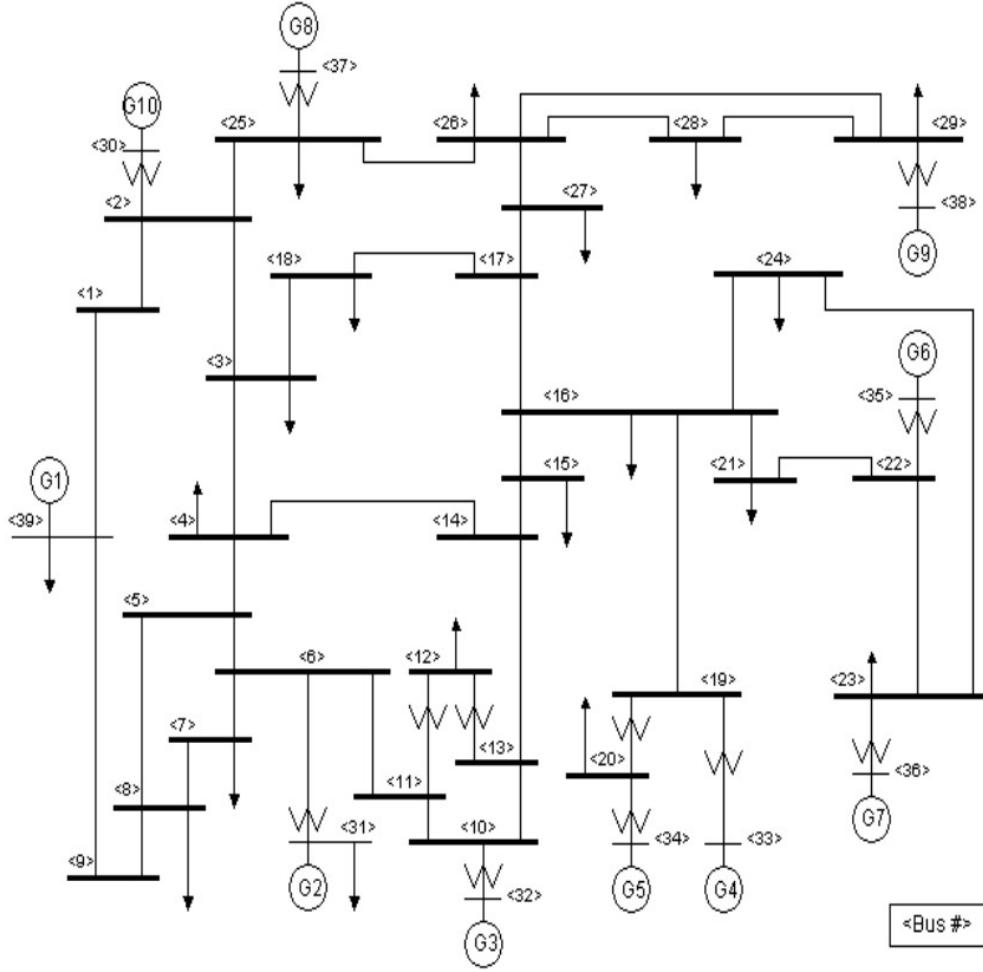


Figure 9: IEEE 39-bus System

To replicate the 39-bus system's topology, we constructed a custom coupling matrix using real 39-bus model line data. In previous simulations, the coupling between nodes was represented by the basic adjacency matrix where:

$$a_{ij} = \begin{cases} 1 & \text{i, j connected} \\ 0 & \text{if i, j not connected} \end{cases} \quad (44)$$

In the realistic IEEE 39-bus simulation, the coupling between busses was derived from the susceptance (the inverse of reactance) between each bus [10].

$$a_{ij} \propto \frac{1}{X_{ij}} \quad (45)$$

The reactance between busses on the grid came from **table**.

4.2.2 IEEE 39 Bus Network Simulations

The Matlab simulation of the IEEE 39-bus system was used to test the effectiveness of the proposed control strategy on a real-world power grid topology. By analyzing the phase and frequency deviations across the IEEE 39-bus system, it is clear that the pinning impulsive control strategy supports the frequency synchronization of power grids.

The Matlab representation of the 39-bus system topology is visualized below in **Figure 10**. Note that the thickness of the lines between nodes (busses) represents the relative coupling between them, which was derived from equation (44).

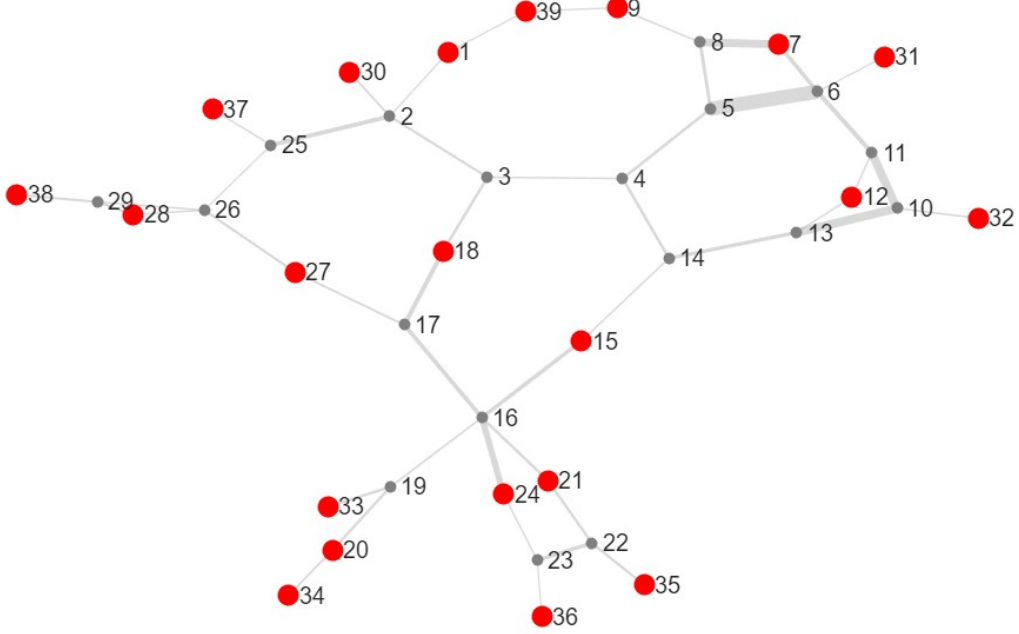


Figure 10: The Matlab Visualization of the IEEE 39-bus System

Real Grid Response to Frequency Distortion

We originally set the state $X = \begin{bmatrix} \Delta\theta \\ \Delta\omega \end{bmatrix}$ to be random deviance from the synchronous state, representing a real grid disturbance. Where X has an index for the phase deviance and an index for frequency deviance for each of the 39 busses (ie. 78 elements array).

Due to the power grid dynamics, the frequency deviance initially spikes. The control method then updates the state of the pinned busses based on the impulse condition until the grid is synchronized.

The speed of synchronization of power grids is essential to the reliability of the power grid. In most standard power grids, there are strict frequency deviation limits. If a generator fails to meet these limits, it must quickly adjust operations to synchronize with the rest of the system.

The **Independent Electricity System Operator(IESO)** manages electricity flow across Ontario. Their Market rules state that any generation facility connected to the system must be able to operate continuously within 59.4-60 Hz [11]. In our state space, this means $\Delta\omega \leq 0.6 \text{ Hz}$ ($\frac{6}{5} \text{ rad/s}$).

To test the real-world performance of the control method, we set the initial frequency to all busses on the system to a random value between 0-2 rad/s (0.32Hz). This high initial system-wide disturbance caused the frequency deviance of one of the generators to spike above the IESO limit of 0.6 Hz. Below, **Figure 11** shows the performance of the control method with this large initially disturbance.

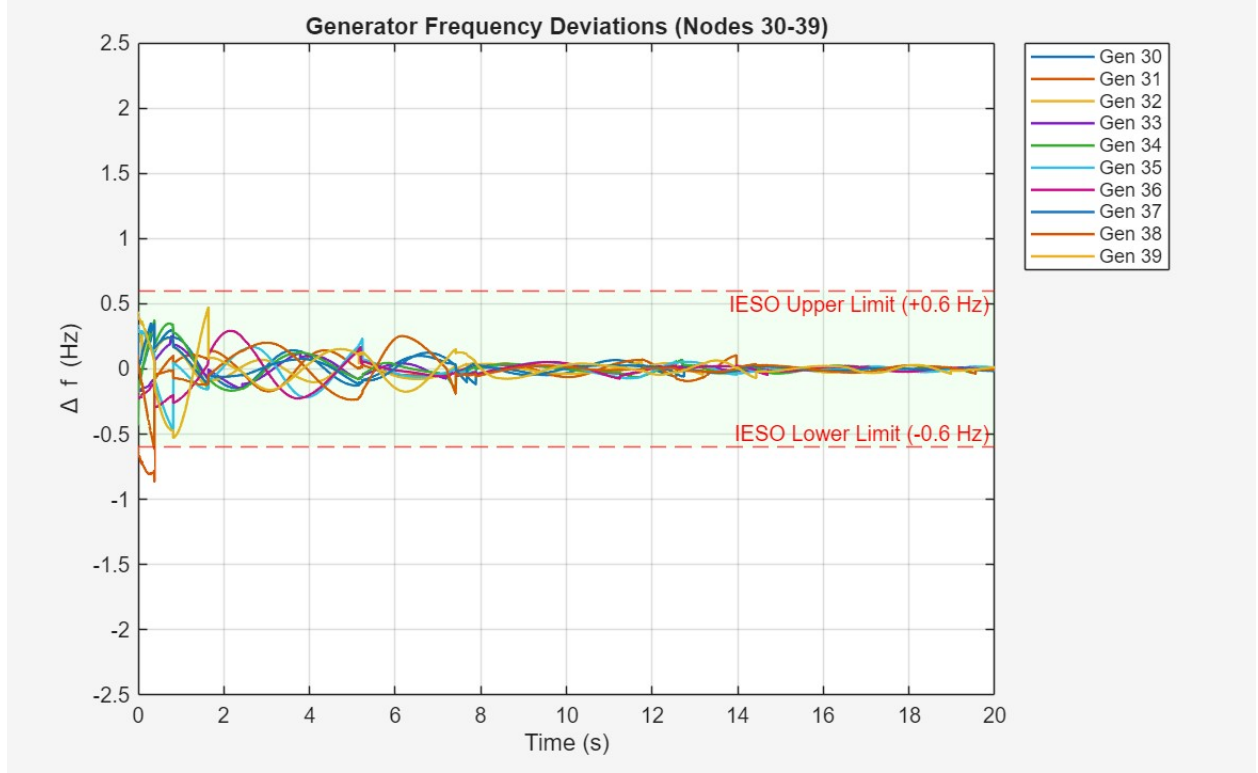


Figure 11: Demonstrates the frequency deviations of the 10 generator busses on the 39-bus system over time

Clearly, the initial disturbance across the grid caused the 10 generators on the system to have a frequency deviation spike. Generator 38 initially spiked to 0.806 Hz, exceeding the IESO threshold. The control method applied impulses to the pinned nodes, resulting in generator 38 transitioning to a deviation within the IESO limit within 0.3798 seconds. This demonstrates that impulsive pinning control strategy can effectively control power grids to meet real-world performance requirements.

Additionally, **Figure 11** demonstrates that frequency reaches a state of synchronization throughout the simulation. The average generator frequency deviation peaked at 0.312Hz right after the initial disturbance, within 20 seconds of the simulation it synchronized to a state with an average generator frequency deviation of 0.0079 Hz.

This shows that even with the high initial system-wide disturbance, the power grid can reach synchronization with the proposed control strategy. Additionally, in this simulation, we are pinning less than half the busses on the system. This indicates that the non-uniform coupling associated with realistic power grids can still achieve frequency synchronization.

4.3 Comparison to Error Based Impulsive Control Scheme

To compare the centrality based pinning algorithm with a different pinning algorithm, an error based impulsive control scheme was simulated. Let N be the maximum number of nodes that can be pinned. The pinning selection is carried out every $T_{interval}$ seconds. Let $e_i(t) = x_i(t) - z(t)$ be the error between node i and the leader at time t . $e_{threshold} \in \mathbb{R}$ is the error value to surpass to be eligible for pinning. Define $E_1 := \{k : |x_k(t) - z(t)| > e_{threshold}\}$, and pin $\max_{u \in E_1} \{|x_u(t) - z(t)|\}$. Then let $E_2 = E_1 \setminus \{u\}$ and pin the max of E_2 , repeating the process until E_n is empty, or $n > N$.

The previous section examined the event-triggered pinning framework within the IEEE-39 bus test case and showed how the controller behaves under a fixed network realization. Having established that simulation setting, the discussion now shifts from individual case behavior to a direct comparison between two

node-selection strategies: the centrality-based pinning method introduced earlier and the error-based pinning method. The purpose of this section is to examine how the choice of pinned nodes affects transient synchronization performance, actuation effort, and convergence behavior as the available actuator budget changes to highlight the advantages and limitations of each strategy.

4.3.1 Comparison Framework

To isolate the effect of node-selection strategy on transient synchronization performance, the error-based and centrality-based pinning methods were simulated under the same linearized second-order Kuramoto dynamics, the same event-triggering law, the same impulsive update rule, and the same initial-condition framework. In both cases, synchronization was evaluated relative to the same synchronous reference, and all controller parameters were held fixed across trials. In particular, the inertia and damping parameters, represented by M and D , were kept fixed throughout all simulations so that differences in transient behavior could not be attributed to changes in the underlying nodal dynamics. Similarly, the event-triggering parameters α_i and β_i were held constant across the different pinning budgets, ensuring that the frequency of impulsive intervention was not altered by retuning the trigger law between runs.

The only distinction between the two strategies is the rule used to determine which nodes receive impulsive control. In the centrality-based case, pinned nodes are selected according to the structural ranking induced by the network topology, whereas in the error-based case, pinned nodes are selected according to the instantaneous magnitude of local synchronization error. This allows the comparison to focus specifically on the effect of structural versus state-responsive node selection, rather than on differences in the controller dynamics themselves.

Performance is evaluated using four main metrics. The aggregate synchronization energy $V(t)$ is used as the primary global measure of how far the network remains from the synchronous manifold over time. The phase trajectories show how rapidly angular deviations collapse across the network, while the frequency trajectories indicate how quickly the nodal frequencies are damped toward a common value. In addition, the impulse-count distributions provide a measure of control activity, showing how often individual pinned nodes are required to intervene during the transient. Finally, convergence time is used to summarize overall performance at the system level by recording how long each strategy requires to drive the network below the prescribed synchronization threshold for a given actuator budget.

By holding the dynamical and triggering parameters fixed and varying only the node-selection rule, the comparison in this section is designed to show whether the observed performance differences arise from where control is applied, rather than from changes in the underlying simulation or event-triggering design.

4.3.2 State Synchronization Response

The state comparison is shown across pinning budgets of 25, 50, 75, and 100 pinned nodes. For each budget, the aggregate synchronization energy of the centrality-based and error-based methods is first compared, followed by the corresponding phase and frequency trajectories. Across all four cases, the error-based strategy produces a more rapid decay in $V(t)$ than the centrality-based strategy, indicating a faster reduction of the global synchronization error. This improvement is most pronounced at lower pinning budgets, where actuator placement matters most strongly. Because the error-based strategy applies the impulses to the nodes that are currently farthest from the synchronization condition, it reduces the dominant transient error more directly. The phase trajectories below show that the error-based controller drives the node angles toward the synchronous manifold over a shorter time horizon, while the frequency trajectories show that it also suppresses frequency mismatch more quickly during the transient.

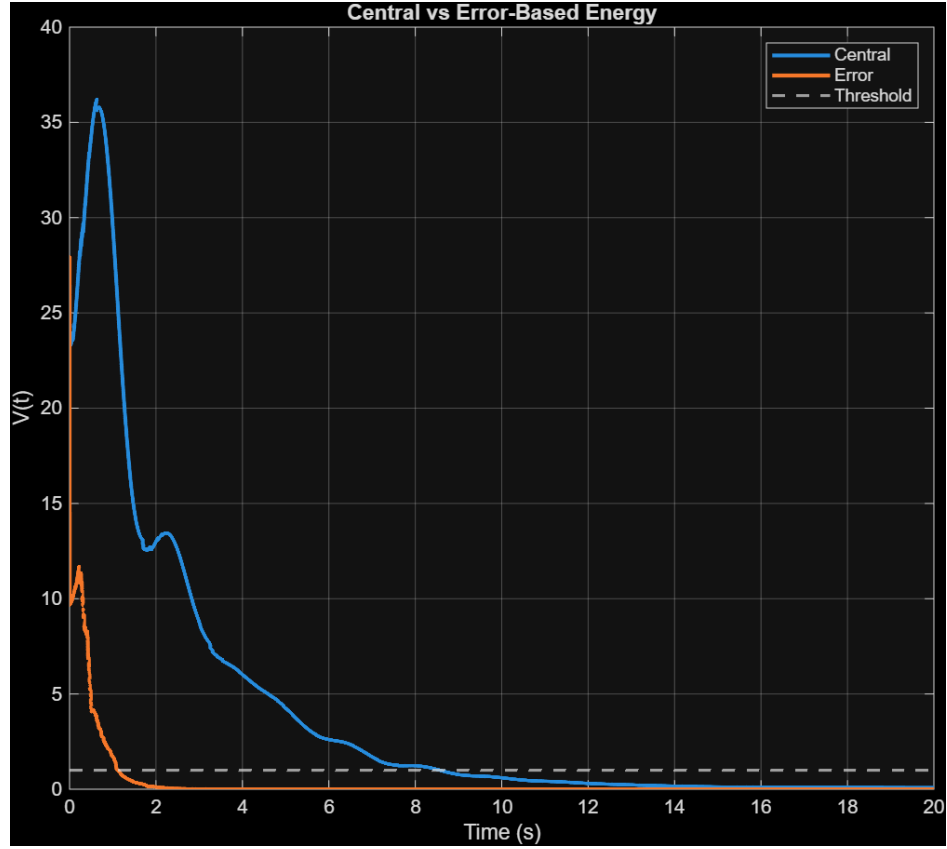
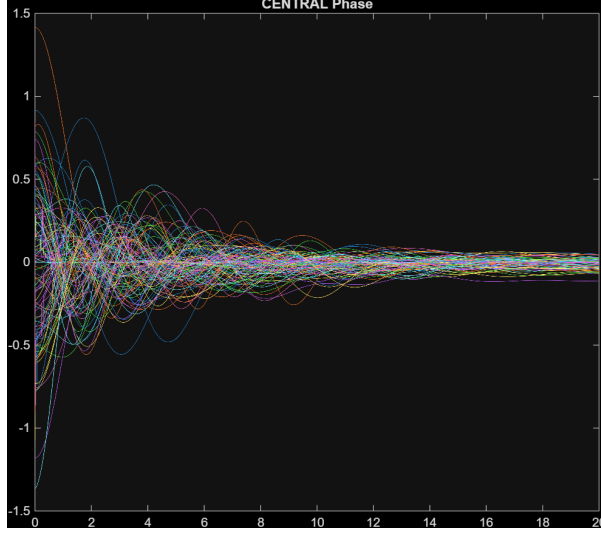
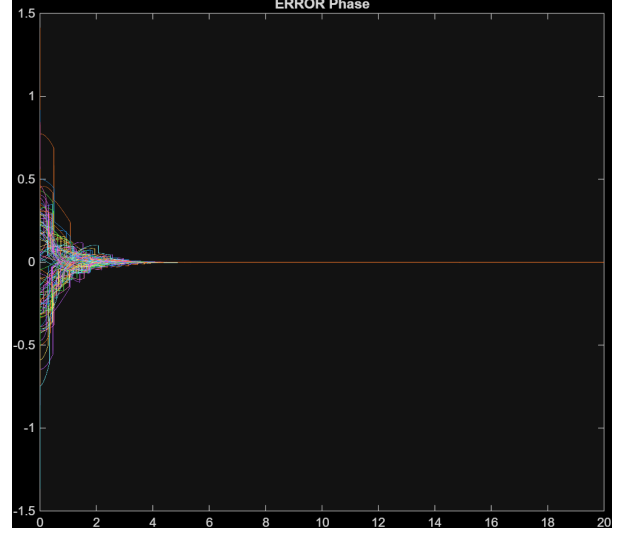


Figure 12: Aggregate synchronization energy for the centrality-based and error-based pinning strategies. The error-based strategy drives $V(t)$ below the convergence threshold more rapidly for the 25 nodes example.

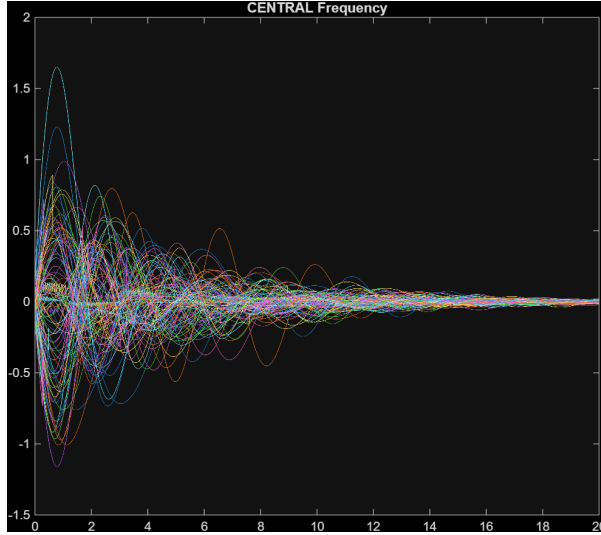


(a) Centrality-based phase trajectories for the 25 nodes example

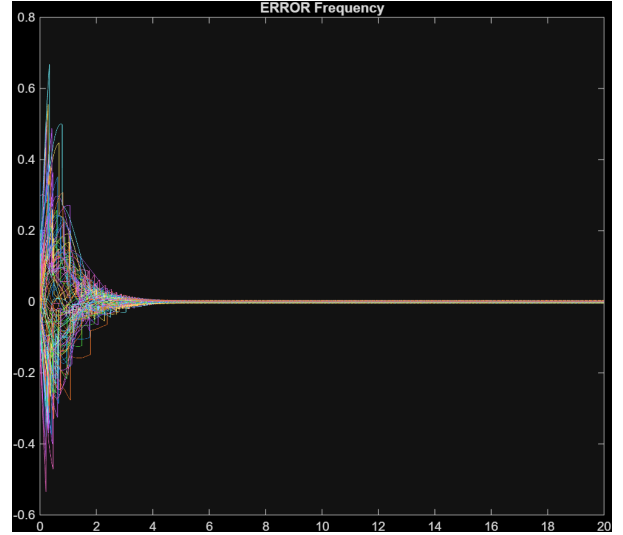


(b) Error-based phase trajectories for the 25 nodes example

Figure 13: Phase trajectories for the 25 nodes-pinned run. The error-based case shows a faster collapse of phase error toward the synchronous manifold than the centrality-based case. Note that the time scale of the error-based method is one fourth of the time scale of the centrality-based method. This gives a better view of the difference in the quantity of impulses.



(a) Centrality-based frequency trajectories



(b) Error-based frequency trajectories

Figure 14: Frequency trajectories for the 25 nodes-pinned run. The error-based case damps frequency mismatch more rapidly than the centrality-based case during the transient.

For the 25-node case, the difference between the two methods is most visible: the error-based method reduces the synchronization energy more steeply and compresses both the phase spread and frequency spread earlier in the simulation.

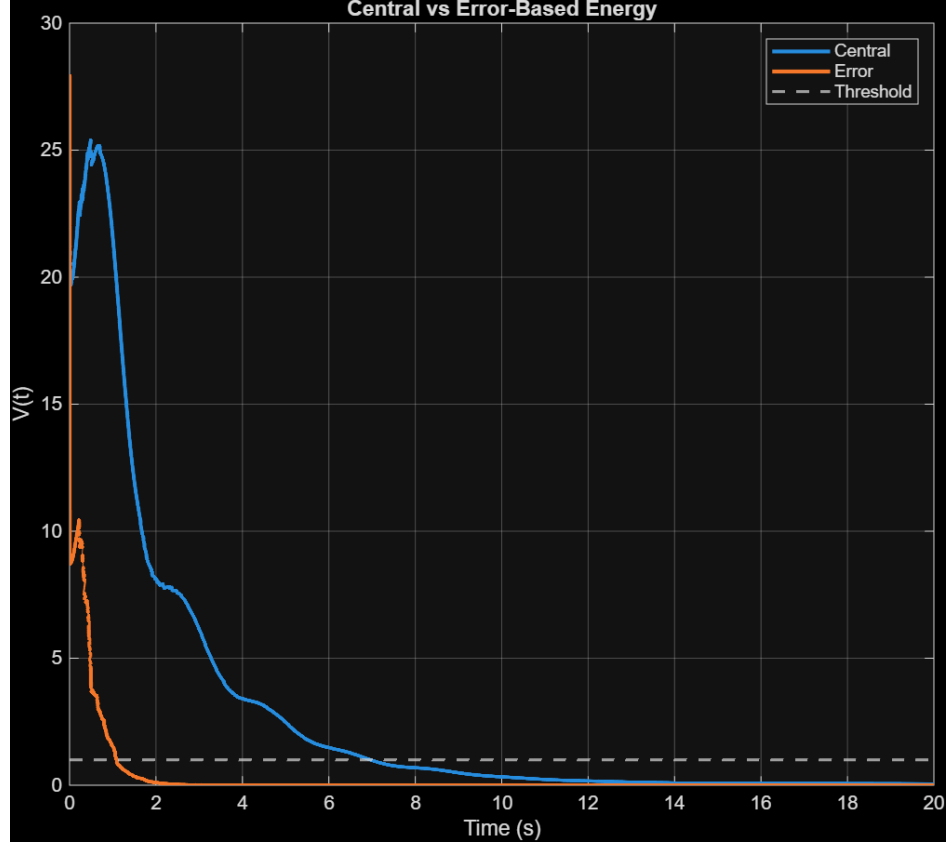
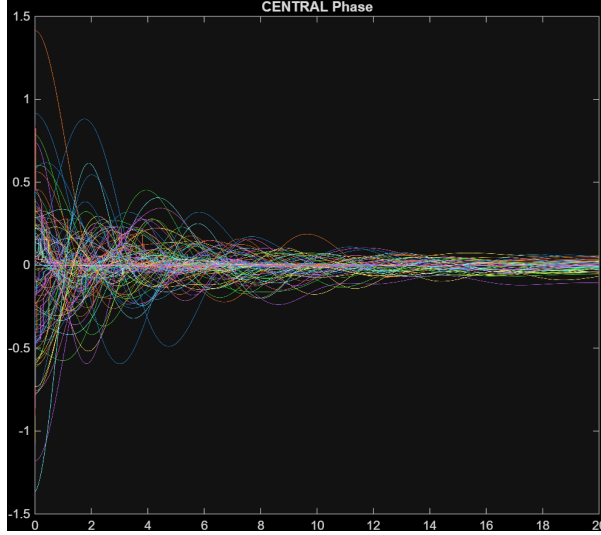
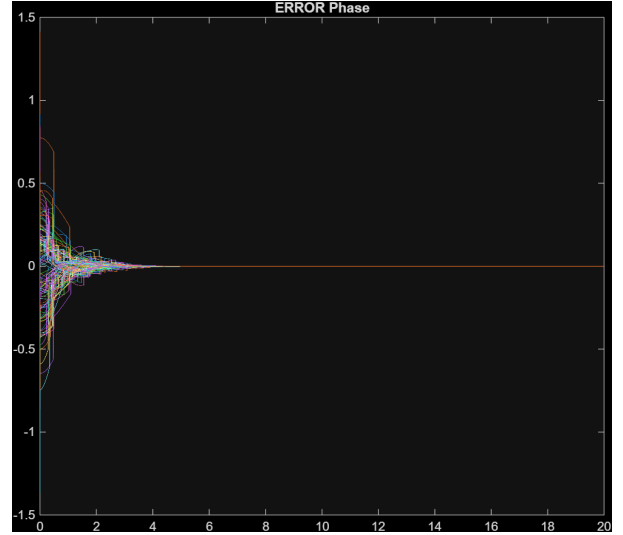


Figure 15: Aggregate synchronization energy for the centrality-based and error-based pinning strategies. The error-based strategy drives $V(t)$ below the convergence threshold more rapidly, however the difference is visibly lower for the 50 nodes example.

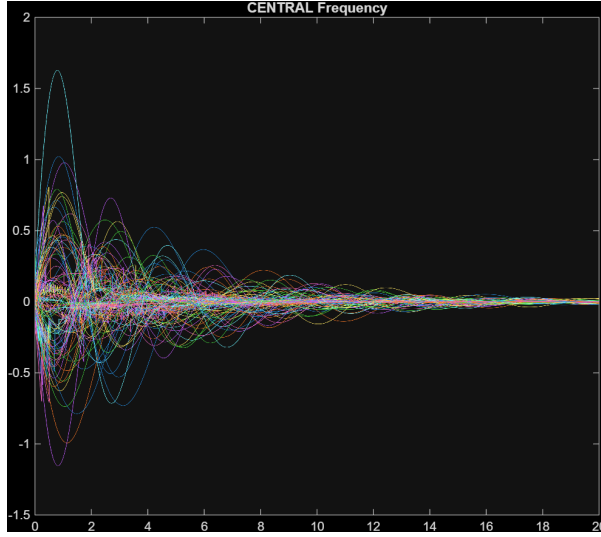


(a) Centrality-based phase trajectories for the 50 nodes example

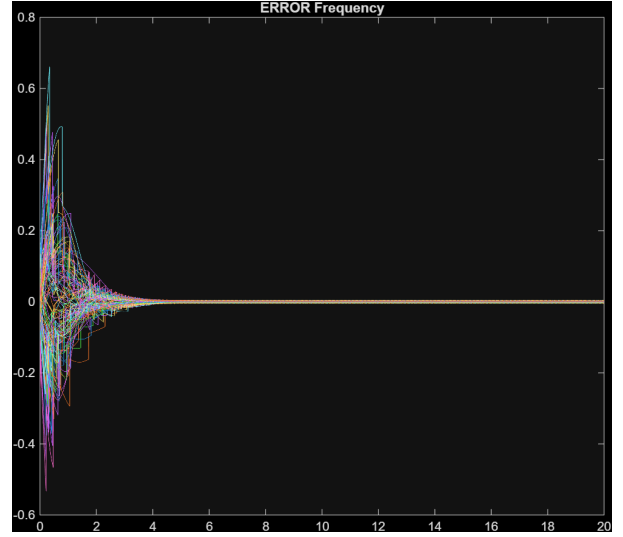


(b) Error-based phase trajectories

Figure 16: Phase trajectories for the 50 nodes-pinned run. The error-based case shows a faster collapse of phase error toward the synchronous manifold than the centrality-based case.



(a) Centrality-based frequency trajectories for the 50 nodes example



(b) Error-based frequency trajectories for the 50 nodes example

Figure 17: Frequency trajectories for the 50 nodes-pinned run. The error-based case damps frequency mismatch more rapidly than the centrality-based case during the transient.

At 50 pinned nodes, the same trend remains, although the gap between the two strategies begins to narrow.

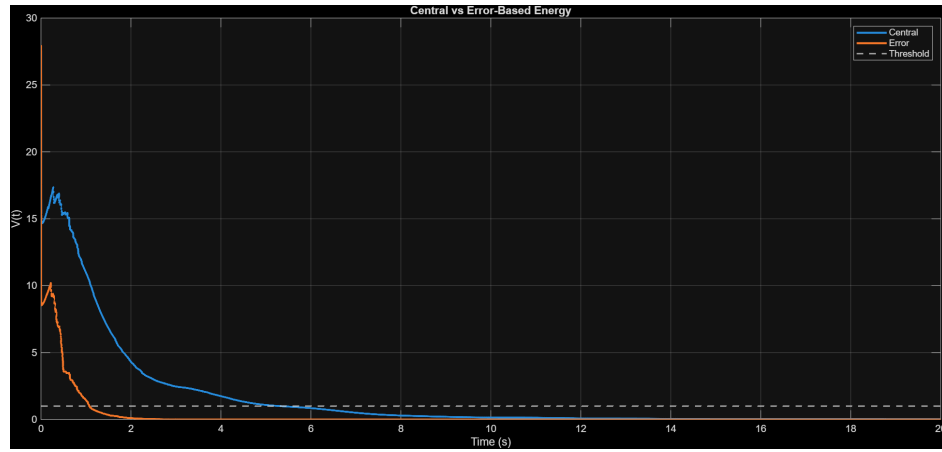
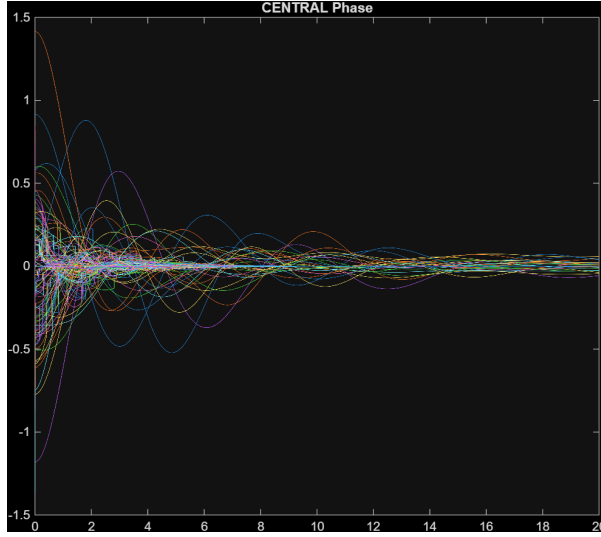
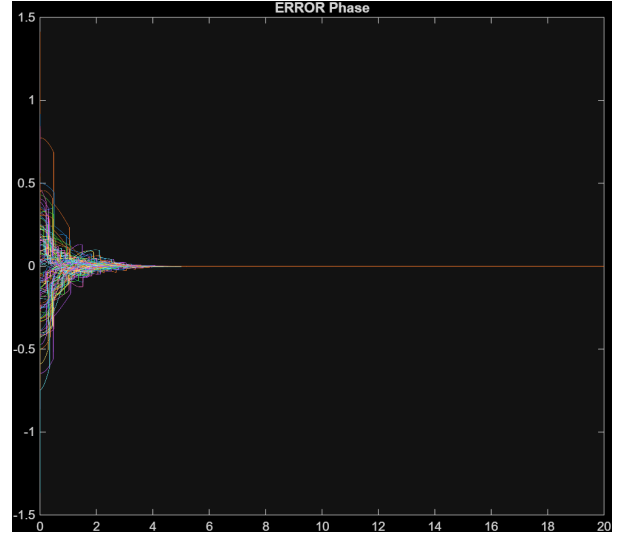


Figure 18: Aggregate synchronization energy for the centrality-based and error-based pinning strategies. The error-based strategy drives $V(t)$ below the convergence threshold more rapidly, however the difference is much lower than it was earlier as seen on the vertical distance between the two curves for the 75 nodes example.

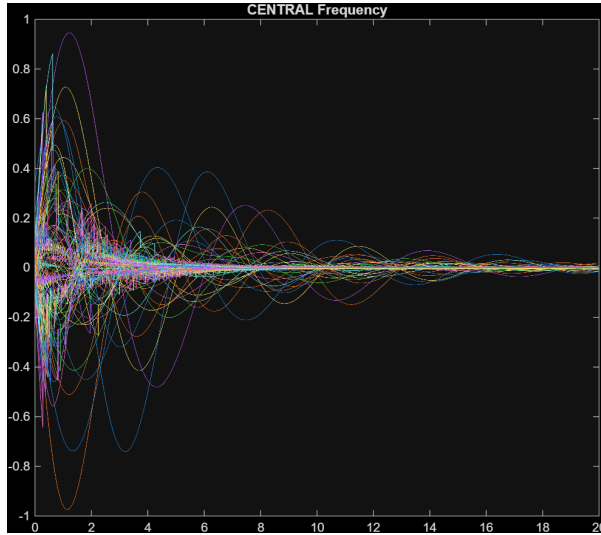


(a) Centrality-based phase trajectories for the 75 nodes example

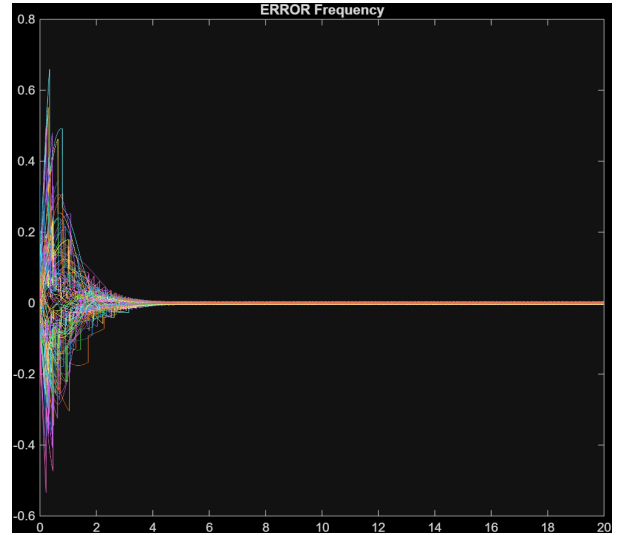


(b) Error-based phase trajectories for the 75 nodes example

Figure 19: Phase trajectories for the 75 nodes-pinned run. The error-based case shows a faster collapse of phase error toward the synchronous manifold than the centrality-based case.



(a) Centrality-based frequency trajectories for the 75 nodes example



(b) Error-based frequency trajectories for the 75 nodes example

Figure 20: Frequency trajectories for the 75 nodes-pinned run. The error-based case damps frequency mismatch more rapidly than the centrality-based case during the transient.

At 75 pinned nodes, both methods exhibit relatively fast convergence, but the error-based method still maintains a visible transient advantage.

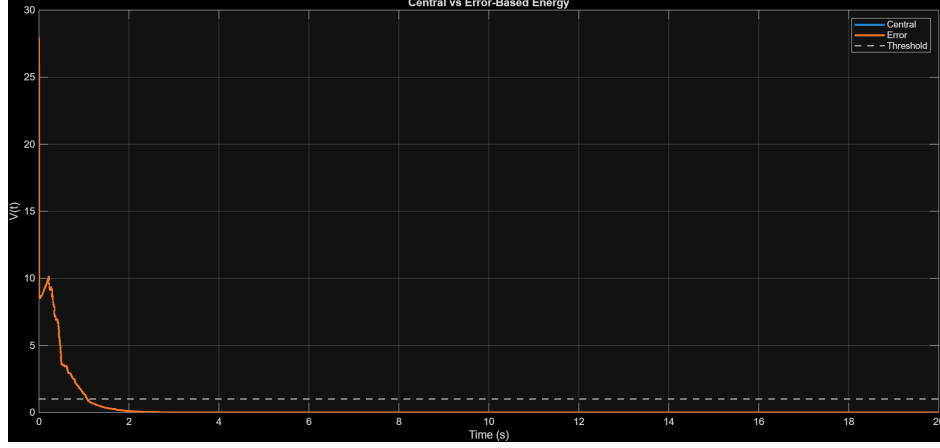
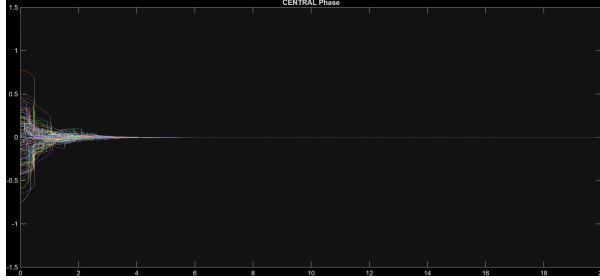
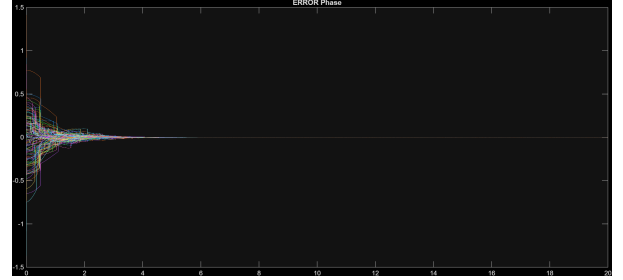


Figure 21: Aggregate synchronization energy for the centrality-based and error-based pinning strategies. The two strategies produce nearly identical energy decay in the fully pinned case, as shown by the overlapping curves.

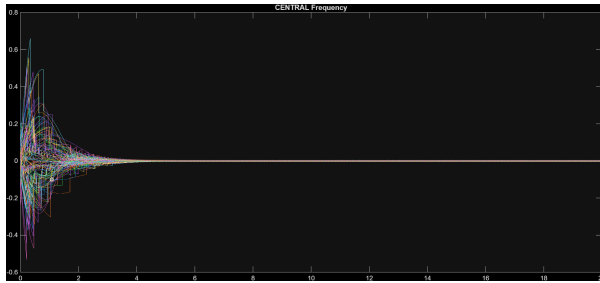


(a) Centrality-based phase trajectories for the 100 nodes example

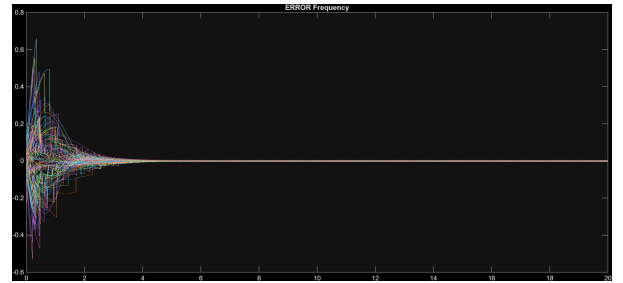


(b) Error-based phase trajectories for the 100 nodes example

Figure 22: Phase trajectories for the 100 nodes-pinned run. The error-based case shows the same rate of collapse of phase error toward the synchronous manifold as the centrality-based case.



(a) Centrality-based frequency trajectories for the 100 nodes example



(b) Error-based frequency trajectories for the 100 nodes example

Figure 23: Frequency trajectories for the 100 nodes-pinned run. The error-based case shows the same rate of collapse of frequency error toward the synchronous manifold as the centrality-based case.

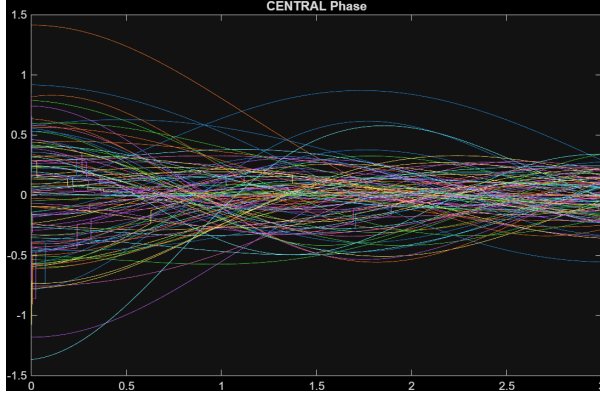
By the fully pinned case of 100 nodes, the distinction becomes much smaller, as both methods now apply control everywhere and therefore operate under nearly identical actuation coverage.

This trend is physically reasonable. The error-based strategy is state-responsive, meaning that it allocates control effort to the nodes that are currently farthest from the synchronous manifold and therefore reacts more aggressively to the evolving transient. The centrality-based strategy, by contrast, is topology-responsive, since it reflects structural node importance rather than instantaneous state deviation. The results therefore suggest that adaptive central-based placement is most beneficial when control authority is limited, and becomes less distinct as the network approaches full pinning.

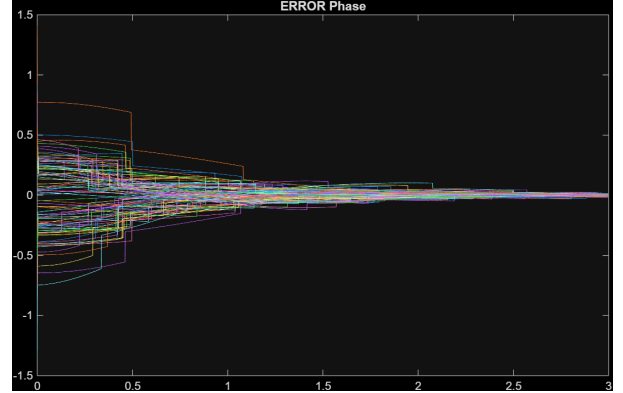
4.3.3 Impulse Activity and Control Effort

The faster convergence of the error-based strategy must be interpreted together with the corresponding impulse-count distributions. For each pinning budget, the centrality-based and error-based impulse counts are plotted separately to show how frequently each pinned node is activated over the course of the simulation. These figures reveal that faster synchronization does not come without cost. In general, the error-based method achieves its improved transient response by directing interventions to the nodes that currently contribute the most strongly to the synchronization error, which directly increases the total actuation burden across a broader portion of the pinned set. The centrality-based strategy, by contrast, may converge more slowly, but it does so using a fixed topology-driven placement rule that does not react to instantaneous state variation. This makes the centrality-based strategy attractive when lower intervention effort is preferred over rapid transient suppression..

To see this difference visually we can shorten the time over which we plot our states. 24 shows the same information as 13, however the shrunk time scale on the error-based phase gives a better visual of the difference in impulse timing and quantity.

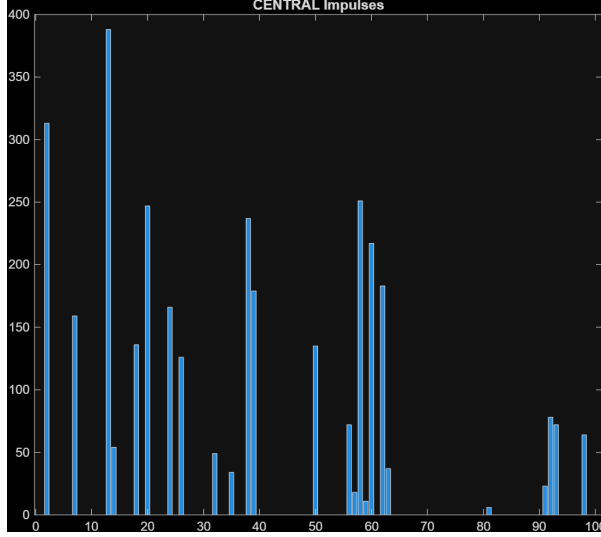


(a) Centrality-based phase trajectories for the 25 nodes example

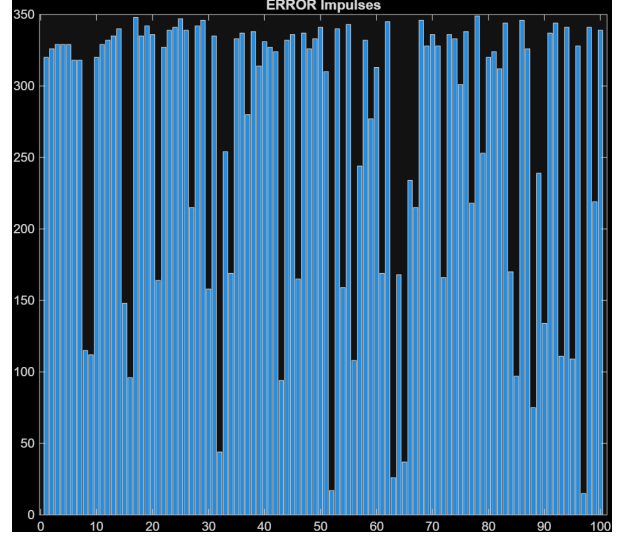


(b) Error-based phase trajectories for the 25 nodes example

Figure 24: Phase trajectories for the 25 nodes-pinned run. The error-based case shows a faster collapse of phase error toward the synchronous manifold than the centrality-based case. Note that the time scale of the error-based method is one sixth of the time scale of the centrality-based method. This gives a better view of the difference in the quantity of impulses.



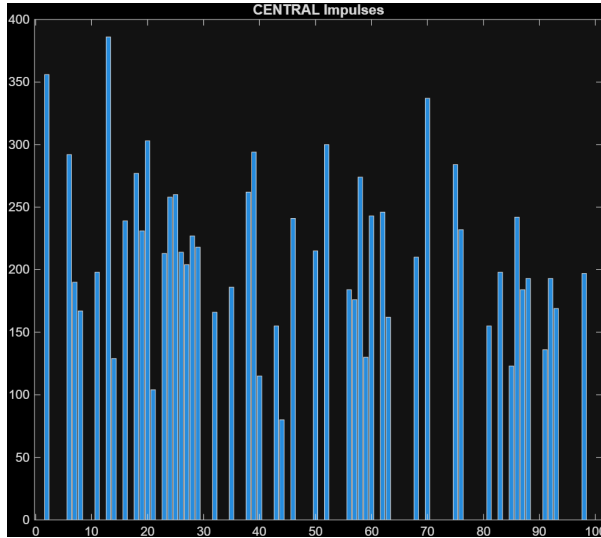
(a) Centrality-based impulse counts at 25 pinned nodes



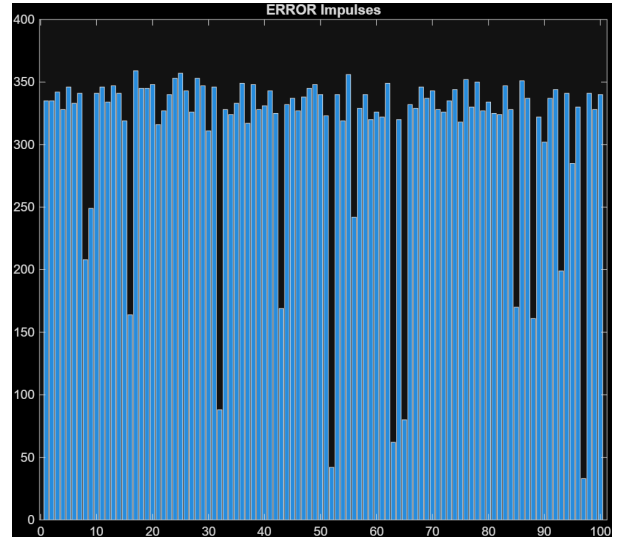
(b) Error-based impulse counts at 25 pinned nodes

Figure 25: Impulse-count distributions for the centrality-based and error-based pinning strategies at 25 pinned nodes. The error-based strategy achieves faster synchronization at this budget, but does so with a noticeably greater actuation burden across the pinned set.

At 25 pinned nodes, the difference between the two strategies is especially informative. The error-based method produces a visibly heavier impulse distribution across the pinned set, indicating that its faster transient response is achieved through more frequent intervention. The centrality-based case, by contrast, shows a lighter overall actuation pattern, which is consistent with its slower but less demanding synchronization behavior. At this low pinning budget, the impulse plots make the tradeoff especially clear: the error-based strategy improves recovery speed, but does so by spending more control effort.



(a) Centrality-based impulse counts at 50 pinned nodes

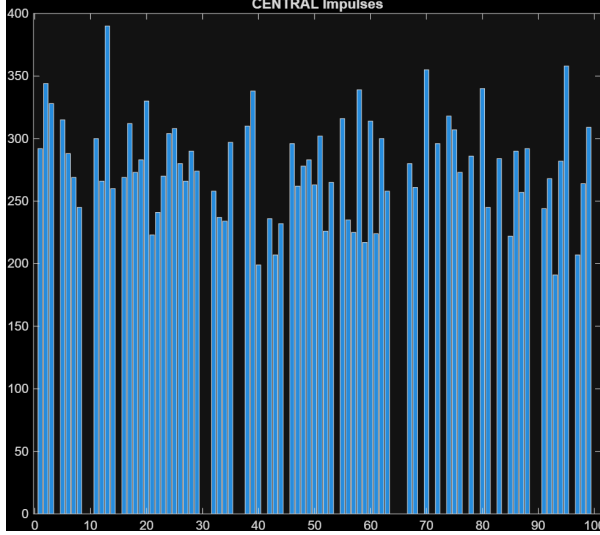


(b) Error-based impulse counts at 50 pinned nodes

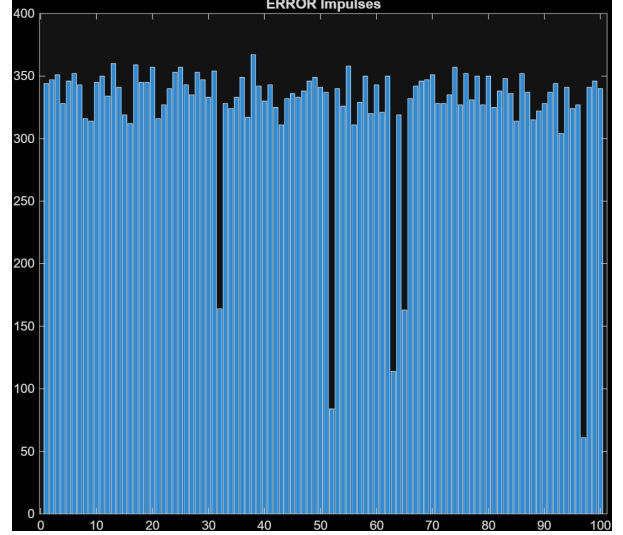
Figure 26: Impulse-count distributions for the centrality-based and error-based pinning strategies at 50 pinned nodes. The error-based method continues to require more frequent intervention, supporting the faster convergence observed in the transient-response comparison.

At 50 pinned nodes, the same pattern remains, although the difference between the two strategies becomes

more distributed across a larger set of pinned nodes. The error-based method continues to show a greater overall impulse burden, reflecting its tendency to repeatedly target nodes with the largest instantaneous deviations. The centrality-based method still requires fewer interventions overall, but this lower actuation cost is accompanied by slower convergence. The comparison at this budget shows that the performance gap observed in the transient-response plots is still supported by a meaningful difference in control effort.



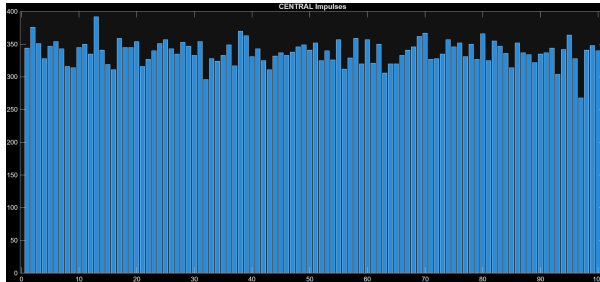
(a) Centrality-based impulse counts at 75 pinned nodes



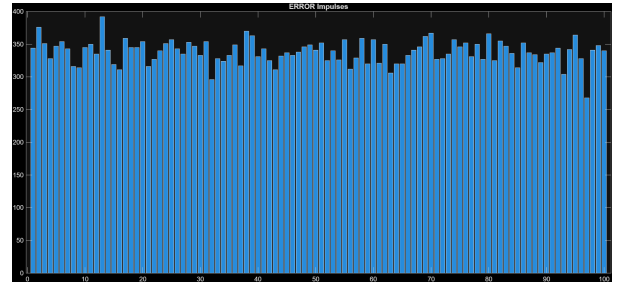
(b) Error-based impulse counts at 75 pinned nodes

Figure 27: Impulse-count distributions for the centrality-based and error-based pinning strategies at 75 pinned nodes. As the pinning budget increases, the difference in impulse activity between the two methods begins to narrow, reflecting the reduced importance of node-selection strategy under broader actuation coverage.

At 75 pinned nodes, both methods exhibit a denser impulse distribution than in the lower-budget cases, since more nodes are available for direct intervention. However, the distinction between the two strategies begins to narrow. The error-based controller still tends to apply impulses more aggressively, but the centrality-based controller now benefits from sufficiently broad placement that its actuation pattern begins to resemble the adaptive case more closely. This result suggests that as the actuator budget grows, the relative importance of adaptive node selection begins to decrease.



(a) Centrality-based impulse counts at 100 pinned nodes



(b) Error-based impulse counts at 100 pinned nodes

Figure 28: Impulse-count distributions for the centrality-based and error-based pinning strategies at 100 pinned nodes. In the fully pinned case, the two strategies exhibit nearly identical impulse behavior, since the node-selection rule no longer differentiates the controller.

In the fully pinned case of 100 nodes, the impulse-count distributions become nearly identical. This is expected, since once all nodes are pinned, both strategies operate under the same triggering condition and

have access to the same actuation coverage. At this point, the node-selection rule no longer differentiates the controllers, and the comparison reduces to two implementations with effectively the same control authority. The impulse plots therefore confirm the same conclusion seen in the transient-response section: the distinction between centrality-based and error-based pinning is strongest when control authority is limited, and weakest when the network is fully pinned.

This trade-off becomes especially important when comparing controller quality from an engineering perspective. If rapid transient recovery is the dominant objective, then the higher actuation burden of the error-based method may be acceptable. If instead the goal is to reduce intervention frequency or simplify implementation, then the centrality-based method may remain attractive despite its slower convergence. In this sense, the impulse-count plots show that convergence speed and intervention burden should not be treated as independent performance measures, but rather as competing design objectives. In the 100-node case, both strategies produce nearly identical impulse behavior, which is expected since all nodes are pinned and the triggering condition is the same in both simulations. Once full pinning is imposed, the distinction between the two node-selection rules disappears.

4.3.4 Effect of Pinning Budget

The overall dependence of performance on actuator budget is summarized by the convergence-time and speed-ratio plots. The convergence-time plot shows how long each strategy requires to drive the network below the prescribed synchronization threshold as the number of pinned nodes increases.

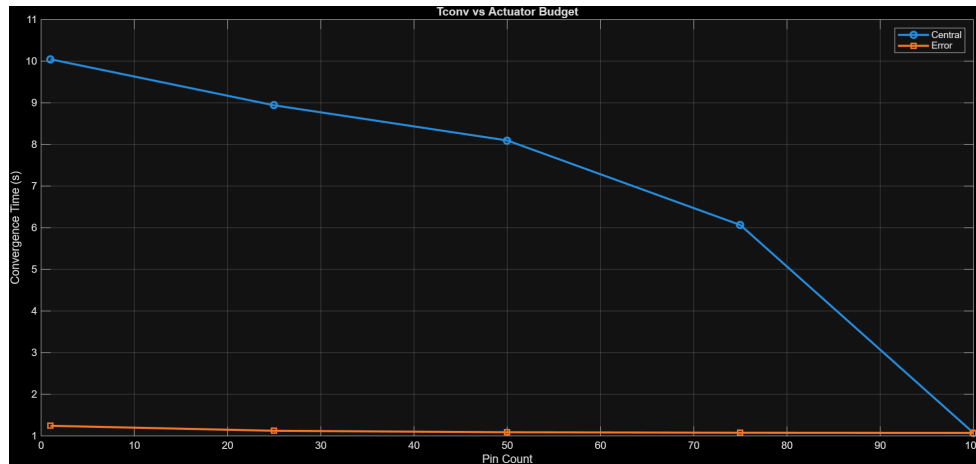


Figure 29: Convergence time as a function of actuator budget for the centrality-based and error-based pinning strategies. The error-based method maintains a substantially lower convergence time across low and moderate pinning budgets, while both methods approach the same behavior in the fully pinned case.

The speed-ratio plot, defined as the ratio of centrality-based convergence time to error-based convergence time, highlights the relative advantage of the error-based method over the full actuator-budget range. At low pinning budgets, the error-based strategy provides the largest improvement, demonstrating that adaptive placement of control effort is most valuable when only a limited number of nodes can be directly actuated. As the number of pinned nodes increases, the difference between the two methods steadily decreases, and the speed ratio approaches unity in the fully pinned limit.

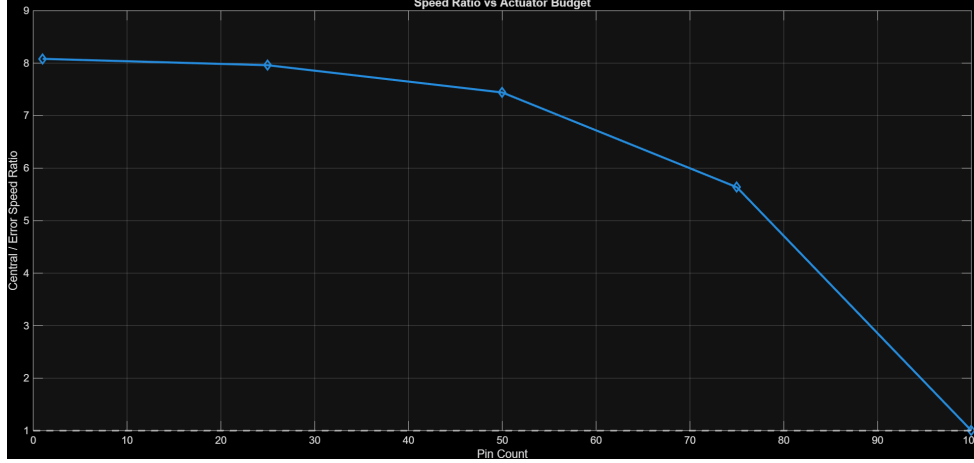


Figure 30: Ratio of centrality-based convergence time to error-based convergence time as a function of actuator budget. The speed advantage of the error-based method is greatest at low pinning budgets and decreases as the number of pinned nodes increases, approaching unity in the fully pinned limit.

In a real network, pinning every node may produce the fastest and most uniform response, but it would generally require the largest sensing, communication, and actuation cost. For that reason, the low- and medium-budget regimes are often the most relevant design cases. The results suggest that error-based pinning offers the strongest benefit precisely in this limited-actuation regime, where central-node-selection quality matters most. At the same time, centrality-based pinning may still provide value as a simpler and more structurally interpretable rule, particularly when a fixed controller placement must be determined offline and updated infrequently. The comparison therefore supports the view that actuator budget should be treated as a primary design variable rather than as a secondary simulation parameter, and that the preferred method depends on whether the design emphasis is placed on fast transient suppression or on softer, more topology-driven deployment.

4.3.5 Error/Central Comparison Results

The information presented shows that error-based node selection reduced the synchronization time, especially when the number of pinned nodes at any given moment is limited. The advantage is supported by the energy, state, and time of convergence plots across all the tested pinning budgets. This improvement comes at the cost of a heavier actuation burden, shown by the impulse-count distribution across the testing pinning budgets. As the number of pinned nodes increases, the difference between both strategies decreases, with the fully pinned cases giving similar results due to the impulse condition being identical. The selection of the applied control comes down to two factors: the speed of convergence required and the available actuation budget. In an application such as power grid stability an argument could be made for both, since the grid requires a low error between states at each node. The error based method would prove useful in situations such as blackouts and disrupted nodes, however the ability to apply a control at every node would create many costs associated with the grid. For a power grid with preset topological features a central-based node selection method would prove less costly, since changes to the system would be rare and the control effort would be lower.

5 Real-Life Application

There is an abundance of pre-existing control methods implemented to maintain power grid synchronization. Our research looks not into replacing any control methods already in use for the power grid, but considers the possibility of leveraging the fast compensation response of inverter based power sources [12].

The diversity of source and load types on a real power grid adds a layer of complexity to the control problem. Event-triggered Impulsive control operates by applying discrete impulses when the triggering

condition is met. The triggering condition would be deviation from the nominal grid frequency of 50/60 Hz [13], but design considerations must be made in order to determine how impulses could be applied to power grid sources. The best candidates for providing impulses would be grid power sources and regulators that are capable of rapidly changing their frequency and phase. Synchronous generators may be a poor fit, due to the inertia associated with the mass of the rotor resisting sudden changes. Renewable energy sources, such as solar, produce DC power that is converted to AC by an inverter. Wind turbines generate power with a synchronous generator, but not at the grid frequency due to the variability of wind speed, and have their power converted to 60 Hz AC by an AC-DC-AC converter (a rectifier and inverter in series) [14]. Backup energy sources like battery banks also make use of an inverter to convert their power to usable 60 Hz AC. Inverters lack the inertia associated with synchronous generators and can change phase and frequency quickly, making them a good candidate for impulsive control.

An event-triggered impulsive pinning control strategy could be implemented as an extra layer of grid security, acting as a seconds-after disturbance control to stabilize the grid and prevent loss of synchronization. As it requires power sources with fast compensation capabilities though, the ability to implement this strategy highly depends on the types of power sources used in a given network of interest. As inverter based power sources become more prevalent in countries prioritizing development of renewable energy sources, this strategy could be implemented in areas with a high proportion of inverter based power sources.

An advantage afforded by the stability results of event triggered impulsive pinning control is that when the convergence criterion is met (reference equation with $\Lambda < 0$), the network will attain convergence by each node on the grid having impulses applied independently. The separate nodes do not need to communicate, and are only responsible for applying their correcting impulses according the triggering criterion. This would save on complexity for the hardware and software to be implemented on-site at each power source. However means the control strategy's performance is not guaranteed for disturbances caused by topology changes in the network, such as connections or generators going down. To compensate for this, it would be required to have a program that can reanalyze the grid after a topology change and rerun the pinning algorithm until the convergence criterion is met. This would incur additional software and hardware costs, and require communication and remote control between the central pinning program and the pinning candidate power sources.

Further work is needed to produce a control strategy that is verifiably able to perform on a real power grid. Our simulations consider a network being solely managed by one control strategy, when in reality, a power grid is controlled by many layers of different control strategies that need to work in harmony together, no controller fighting against any other, and controllers must have protections in place to prevent them from executing control actions that could harm grid components. Consequently, the true control problem to be solved in order to implement pinning control on a power grid is going to be very different from the one our current simulations have considered. A real-life application is quite far away, and further work is needed

5.1 Triple Bottom Line and Impact of Engineering

A useful way to evaluate the broader significance of the proposed control strategy is through a triple bottom line lens. In this project, event-triggered pinning impulsive control should not be judged only by whether it produces mathematical synchronization, but also by whether it improves grid performance in a way that is socially valuable, environmentally responsible, economically justified, and compatible with real standards and operating constraints [15], [16]. This broader framing is consistent with the presentation section on *Impact of Engineering*, which evaluates the method through social, environmental, ethical, economic, and codes-and-standards considerations rather than through stability theory alone.

The relevance of this broader framing is clear from the overall thesis motivation. Modern power grids can be modeled as large networks of coupled oscillators, and synchronization failures can contribute to instability and cascading outages [17], [18]. The project therefore treats pinning impulsive control not simply as a mathematical stabilization tool, but as a possible sparse corrective layer for physically meaningful power-network disturbances.

5.1.1 Design Problem and Implementation Context

The engineering design problem in this thesis is not simply to prove synchronization of a networked dynamical system, but to improve synchronization performance under realistic constraints on actuation, communication, sensing, and computational burden. Earlier sections show that event-triggered pinning impulsive control can improve recovery in linearized power-grid-like networks while avoiding unnecessary intervention by acting only at selected nodes and only when a local threshold condition is violated [19]. This sparse structure is important because a large real-world grid cannot be treated as though every node can be continuously monitored and directly controlled with no cost or implementation burden.

This design objective is constrained by the fact that performance, simplicity, and realism do not improve together automatically. The simulation comparison between centrality-based and error-based pinning shows that faster convergence can be achieved when impulses are directed toward the nodes with the largest instantaneous local error, especially when only a limited number of nodes are pinned. However, the same comparison shows that this faster convergence is accompanied by heavier impulse activity and therefore greater control effort. The design problem is therefore not merely to find the fastest controller, but to identify a control strategy that provides acceptable disturbance recovery while respecting practical limits on intervention frequency, implementation complexity, and operational cost.

There are also important implementation constraints. The simulations idealize the controller by assuming that nodes are broadly pinnable, that impulses can be applied instantaneously, and that the dynamics remain well represented by a linearized second-order Kuramoto or swing-like model near the synchronized state [18], [19]. In practice, the controllability of nodes is highly uneven. Generator buses, inverter-based resources, storage devices, and passive loads do not all have the same ability to provide rapid corrective action. Similarly, the model assumptions become less reliable as the system moves farther from the operating point around which the equations were linearized. For that reason, the present results are most appropriately interpreted as proof-of-concept evidence for a sparse corrective architecture, rather than as field-ready performance claims.

5.1.2 Social Impact

The strongest social argument for the proposed method is improved grid reliability during disturbances. The slide set explicitly states that grid reliability protects critical infrastructure, including transportation, communications, and health care, from catastrophic failure, and that improved post-disturbance recovery strategies are important for mitigating outage implications [20]. This is not a minor side benefit: modern social and economic life depends on the continuous operation of systems whose basic functionality collapses when electricity becomes unavailable. A method that shortens the time required to restore synchronization after a disturbance therefore has social value because it may reduce the likelihood, duration, or severity of interruption to essential services.

This social importance becomes even stronger when viewed through the lens of unequal vulnerability. The presentation notes that power outages can have disproportionate impacts on vulnerable communities and that, for communities with inadequate resources for backup power, efficient disturbance recovery is especially critical [20]. This point should be stated explicitly in the thesis because it elevates the discussion beyond generic reliability language. In many communities, backup generators, energy storage, redundant services, or rapid emergency response are not equally available. A prolonged outage can therefore create unequal harm. Households with medical devices, low-income households without access to alternative heating or cooling, remote communities, and public-facing services with limited resiliency are often less able to absorb disruption. In such settings, even a modest reduction in outage duration or post-disturbance instability can have ethically and socially meaningful consequences.

At the same time, the social benefits of the method should not be overstated. The present thesis does not prove that the controller would eliminate blackouts or guarantee service continuity for all users. What it does show is that a sparse corrective layer can improve synchronization recovery in the modeled setting. The socially relevant conclusion is therefore that faster disturbance recovery may contribute to better resilience for critical infrastructure and for communities that lack backup power, not that the controller itself solves the broader reliability problem.

5.1.3 Environmental Impact

The environmental case for event-triggered pinning impulsive control must be made carefully, but it is still meaningful. The slide deck argues that pinning impulsive control minimizes energy spend on controlled nodes and that improved synchronization may support the integration of renewable energy sources [21]. These claims should be interpreted in a systems sense rather than as a direct emissions accounting result.

The first environmental advantage comes from the nature of impulsive control itself. A continuous full-effort control architecture applies corrective action all the time, including when the system is already close to acceptable operation. By contrast, an event-triggered impulsive controller acts only when a threshold is violated and only at the selected pinned nodes [19]. This means that, at least in principle, the controller avoids unnecessary sustained correction effort and reduces the amount of active intervention imposed on the participating assets. In practical terms, this can mean lower control energy expenditure, less persistent command activity, and less reliance on continuously active actuation when compared with a broader continuous-control strategy. The environmental relevance of this point is not that the impulses are free, but that they are sparse and conditional rather than continuously applied across the network.

The second environmental argument is indirect but important. If synchronization can be maintained or restored more effectively in systems with high shares of fast-response, inverter-based resources, then the method may support the reliable integration of renewable energy sources [16], [21]. This argument is plausible because renewable-rich systems often depend more heavily on power-electronic interfaces and faster control layers than traditional synchronous-generation systems. A sparse, fast-response corrective strategy may therefore help stabilize operating regimes that are increasingly relevant in cleaner electricity systems.

However, these environmental claims have clear limitations. The thesis does not directly quantify emissions reduction, does not perform a life-cycle analysis, and does not prove that total grid energy consumption decreases because of the controller. It would therefore be too strong to claim a direct environmental benefit in the same way one would claim reduced fuel use or reduced carbon intensity. The more defensible conclusion is that event-triggered impulses may be environmentally preferable to continuous corrective effort because they reduce unnecessary control activity, and that improved synchronization capability may indirectly support cleaner grid operation by making renewable-rich systems easier to stabilize.

There are also potential environmental downsides to acknowledge. A more aggressive error-based strategy applies more frequent impulses, which may increase cycling or stress on the assets that provide correction. If those assets are batteries, inverter interfaces, or other electronically controlled equipment, heavier intervention could contribute to wear, maintenance burden, or shortened service intervals. From a sustainability perspective, this means that the environmental value of the controller depends not only on faster convergence, but also on how much corrective effort is demanded from the participating hardware.

5.1.4 Ethical and Equity Considerations

The ethical dimension of this project is closely connected to the public necessity of electricity. A control method for grid stability cannot be judged purely by mathematical elegance; it must be judged by whether it supports safe and equitable operation of infrastructure that large populations depend on. The first ethical consideration is therefore safety. Because unstable frequency and loss of synchronization can damage equipment and contribute to cascading outages, any future implementation would need to prioritize safe and reliable operation over aggressive performance gains [15], [22]. A controller that improves convergence in simulation but interacts poorly with protection or plant-level controls would not be ethically acceptable in a real grid.

A second ethical issue is equity of impact. As noted above, outages do not affect all users equally. Communities with little or no backup power are more exposed to the consequences of instability, especially when service interruptions affect heating, cooling, refrigeration, medical devices, communications, or public services [20]. In that sense, disturbance recovery has an equity dimension. A supplementary control strategy that reduces recovery time can provide disproportionate benefit to the users who are least able to tolerate service interruption. At the same time, deployment choices could themselves create inequity if enhanced control infrastructure is concentrated only in regions or assets that already have better resilience. Ethical engineering design would therefore need to consider not only whether the controller works, but also who benefits from it and under what deployment assumptions.

There is also an ethical responsibility to communicate model limits honestly. The present work relies on a linearized model, idealized instantaneous impulses, and simplified controllability assumptions [18], [19]. These are reasonable choices for a thesis at the intersection of control theory and engineering application, but they are still limitations. Ethical engineering practice requires that these limits be disclosed clearly so that the method is presented as a promising concept with identifiable boundaries, not as a complete and immediately deployable solution.

5.1.5 Economic Impact and Feasibility

The economic rationale for event-triggered pinning control comes from the same design idea that motivates it technically: not every node must be directly controlled at all times. By applying corrective action only at selected nodes and only when the local error exceeds a threshold, the method has the potential to reduce implementation burden relative to broader continuous-control architectures [19]. In principle, this can reduce sensing, communication, and actuation requirements, which is especially important in geographically large and operationally complex power networks.

The slide deck also connects grid disturbance recovery directly to financial consequences by citing significant outage costs in Canada [23]. That framing is useful because it turns synchronization improvement into an economic question rather than a purely dynamical one. If a sparse corrective controller can reduce the severity or duration of disturbance-driven instability, then its value may lie partly in avoided outage costs, avoided equipment stress, and avoided operational disruption.

However, the economic case depends strongly on implementation structure. A sparse controller is only economically attractive if the infrastructure needed to support it remains justified by the performance benefit it provides. In practice, a deployable architecture would still require synchronized measurement, some communication layer, plant-level integration at controllable assets, and supervisory logic for updating or validating pinning decisions [15], [16]. The most plausible economic interpretation is therefore not that the controller is inherently cheap, but that it may provide better value than a denser or continuously active control scheme if it delivers enough disturbance-recovery improvement per unit of added infrastructure.

5.1.6 Economic Trade-Off Between Centrality-Based and Error-Based Pinning

The comparison between centrality-based and error-based pinning sharpens the economic analysis because the two strategies imply different cost structures. The simulation slides show that error-based pinning provides the largest speed advantage at low pinning budgets and that both methods converge faster as the number of pinned nodes increases. They also state explicitly that error-based pinning is associated with faster convergence and higher intervention burden, while centrality-based pinning is associated with slower convergence and simpler fixed deployment. This is a crucial engineering trade-off.

A centrality-based deployment is likely to be cheaper and easier to maintain because the pinned set can be determined offline from network topology and then updated only when the system structure changes [17]. That reduces the need for continuous ranking of node error, lowers supervisory decision complexity, and may reduce the amount of real-time data processing required during operation. In practical terms, this means the communication, monitoring, and algorithmic layers can be simpler. From a lifecycle perspective, that simplicity may translate into lower integration cost, lower validation burden, and easier explainability for operators.

Error-based pinning, by contrast, is more adaptive but likely more expensive to operate. Because the pinned set is recomputed continuously from instantaneous local error, the controller demands more frequent state evaluation, more real-time supervisory logic, and more active decision-making. The same simulation results also show heavier impulse activity in the low- and mid-budget regimes, which suggests greater actuation demand on the participating assets. This is important economically because faster convergence is not free: it may require more measurement bandwidth, more online computation, more control-system validation, and potentially more equipment wear or maintenance burden.

The economically preferred method therefore depends on what cost dominates in the intended application. If the largest cost is the consequence of poor or slow disturbance recovery, then the additional complexity of error-based pinning may be justified because it provides the strongest benefit precisely when control resources are limited. If, however, the dominant cost is ongoing monitoring, integration, validation, and maintenance burden, then a simpler centrality-based scheme may offer better lifecycle value even if it is not the fastest

option. This means the comparison cannot be reduced to “which controller converges faster.” The more appropriate engineering question is which strategy provides the best performance per unit of implementation burden.

5.1.7 Codes, Standards, and Professional Considerations

The presentation explicitly connects the project to real power-system operating requirements by citing the Ontario requirement that generation and storage facilities must operate between 59.4 Hz and 60.6 Hz [24]. This is valuable because it gives practical meaning to the recovery results shown in the simulation section. A controller that restores generator frequency back within that band can be compared against a real operating requirement rather than only against an arbitrary simulation threshold.

The slide set also references IEEE 1547 as a relevant interconnection and interoperability standard [16]. This is especially important because inverter-based resources are the most plausible fast-response candidates for future pinning implementations. More broadly, any realistic deployment would need to operate within a standards and compliance environment that already governs connected grid assets. As discussed in the existing write-up, the IESO framework requires performance modeling, data provision, monitoring capability, and approval for changes that may affect reliable operation [15]. Likewise, NERC protection requirements constrain how generating resources may respond during disturbances [22]. A supplementary impulsive controller could therefore not be justified solely because it improves simulated synchronization. It would have to coexist safely with ride-through requirements, protection settings, plant-level controls, and operator approval processes.

From a professionalism standpoint, this means the design problem extends beyond control theory. A practical implementation would require validated trigger logic, auditable measurements, clearly defined operating envelopes, interoperability with existing equipment, and a credible integration path under the standards environment already governing grid assets. The professional question is therefore not simply whether the controller works mathematically, but whether it can be deployed in a way that is compliant, testable, and safe.

5.1.8 Trade-Offs Among Performance, Cost, and Scalability

The clearest engineering trade-off demonstrated in the thesis is between convergence speed and control effort. Error-based pinning improves synchronization most strongly when control authority is limited, but this improvement comes with heavier impulse activity. Centrality-based pinning is slower, but more structurally interpretable and easier to deploy as a fixed rule. A second trade-off exists between breadth of pinning and implementation cost. As the number of pinned nodes increases, the difference between the two strategies becomes much smaller, and nearly disappears when the network is fully pinned. Full pinning may therefore appear attractive from a purely control perspective, but it would also imply the greatest sensing, communication, and actuation burden.

This is why the low- and medium-budget regimes are the most practically important. They force the designer to decide how limited control authority should be allocated, how much monitoring and actuation burden is acceptable, and what recovery improvement is worth paying for. In that sense, the simulations do more than compare two algorithms: they identify the operating region where node-selection strategy has the most practical value.

5.1.9 Evaluation of Information Sources

The evidence used in this section comes from several source types, and they do not all serve the same purpose. Theoretical claims about event-triggered pinning, sparse impulsive control, and synchronization guarantees are grounded in the control literature used throughout the thesis, especially the event-triggered pinning paper that motivates the local triggering structure and sparse intervention logic [19]. Application-oriented claims rely more heavily on the thesis simulations and on operational references such as IESO requirements and IEEE interconnection standards [15], [16], [24]. Broader societal and economic claims are supported by infrastructure and outage-impact sources [20], [23].

This distinction matters for credibility. Theory justifies the control concept, simulations illustrate its behavior under the chosen assumptions, and regulatory or application sources explain what would matter

in practice. The most responsible interpretation of the evidence is therefore layered: the controller is theoretically motivated, numerically promising, and practically relevant, but still bounded by modeling assumptions and real implementation constraints.

6 Opportunity for Future Work

This document demonstrates that the pinning impulsive control strategy is a viable option for maintaining grid synchronization. This section will introduce future work to refine the control strategy for realistic grid control.

6.0.1 Verifying stability of frequency only impulsive pinning control

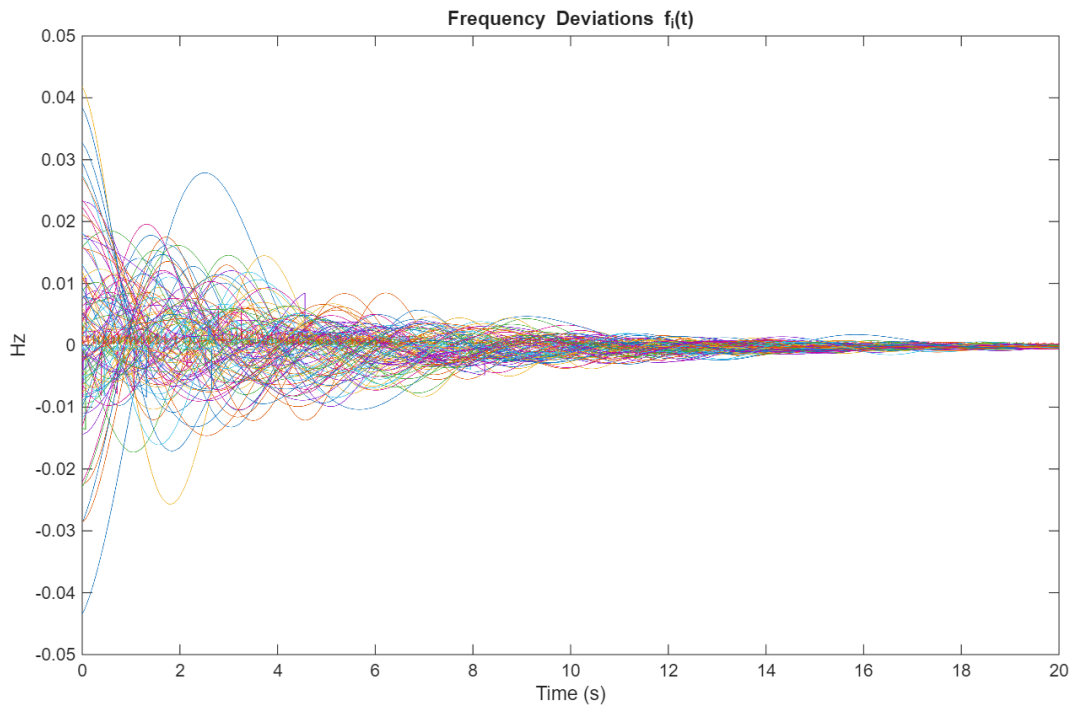


Figure 31: Caption

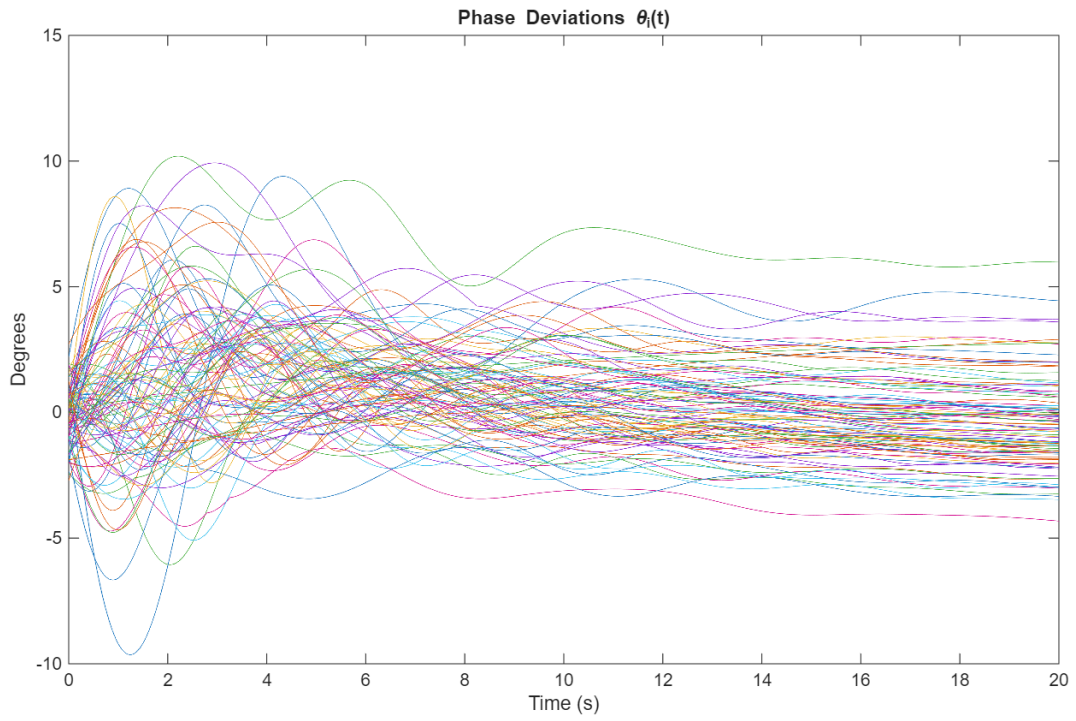


Figure 32: Caption

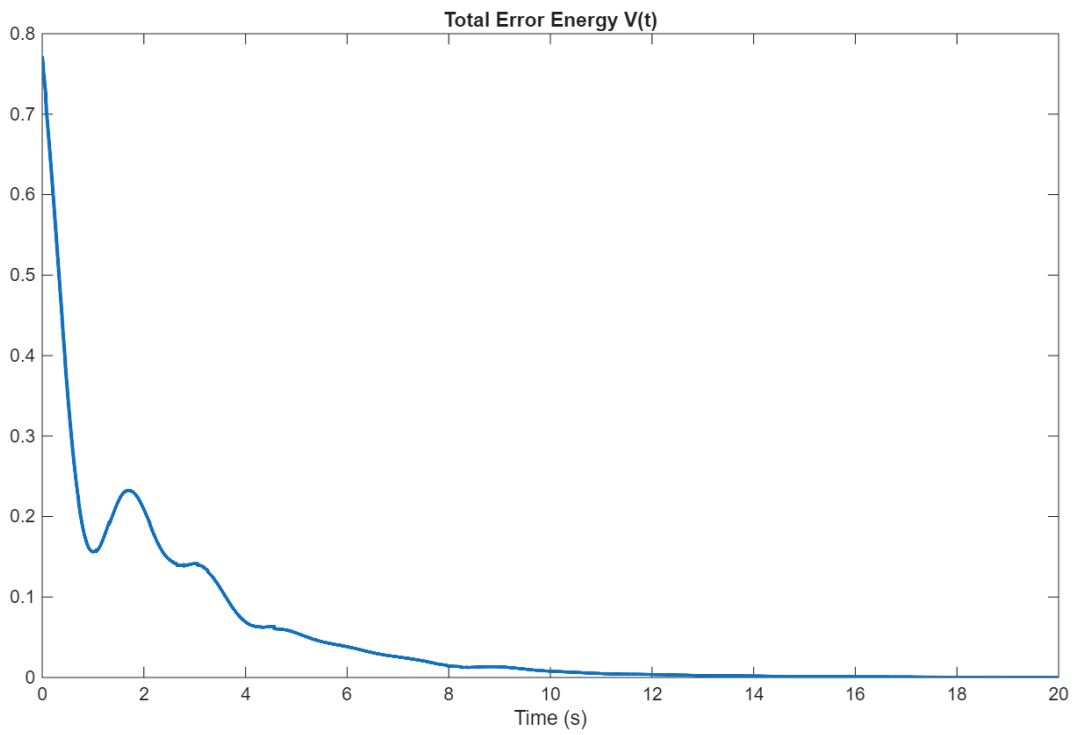


Figure 33: Caption

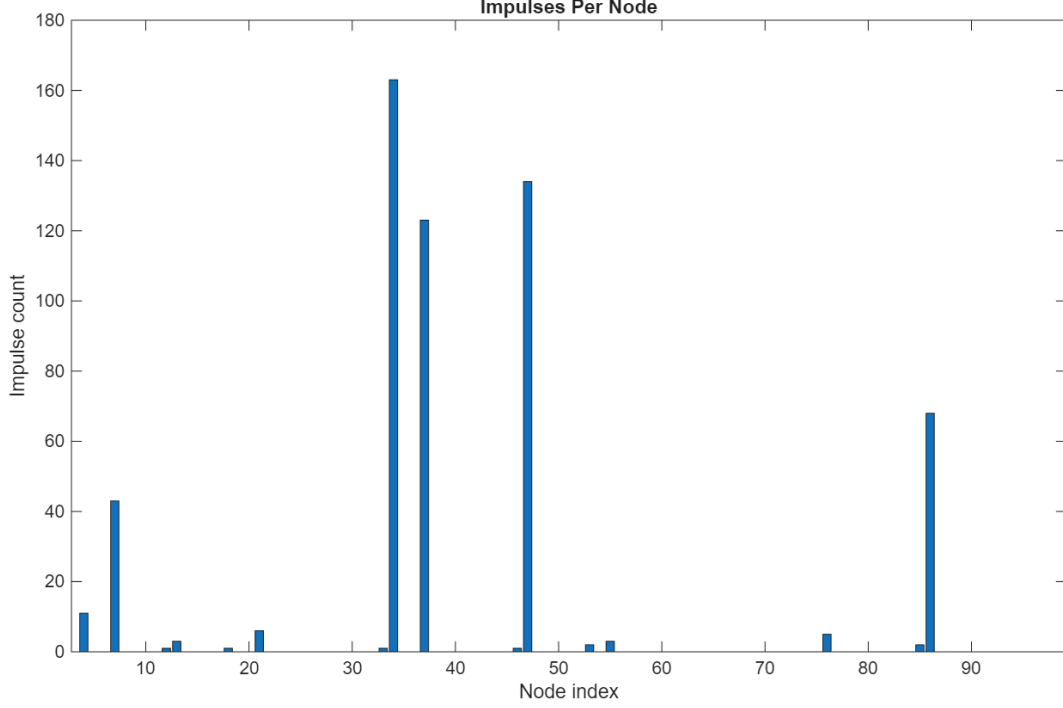


Figure 34: Caption

6.1 Renewable-Prioritized Pinning Strategies

The simulations discussed previously used centrality based pinning and error based pinning strategies, each of which had their own benefits and drawbacks. Instead of exclusively relying on one of these pinning rules, future iterations could test a pinning strategy that applies old pinning strategies along with a generator type weighting. This strategy would identify the type of generator for each generator node on the network and prioritize pinning renewable generators on the network. Specifically, assigning higher weights to inverter-based generators such as wind and solar facilities. Inverter-based generators lack the high inertia associated with synchronous generators, allowing for quick phase and frequency changes required for effective event-triggered control.

The new pinning algorithm could implement previous pinning strategies while minimizing a generator type cost function, such as:

$$C = \sum_{i=1}^N \|e_i\|^2 + \lambda(\text{Number of Non-Renewable Pinned Generators}) \quad (46)$$

Such a pinning strategy could improve grid control algorithms by taking advantage of the high-speed frequency updates of inverter-based generators.

6.2 Variable Natural Frequencies

In the linear simulations discussed so far, we assume uniform natural frequency for each node across the network. In a realistic power grid, load imbalances may cause different areas of the grid to exhibit different frequencies based to their natural frequency differences.

In the future iterations, it would be valuable to simulate the impulsive pinning control strategy with the natural frequency $\omega_i^{natural}$ varying for each node. Analyzing the performance of the control strategy when each node wants to oscillate at its own natural frequency would provide a more rigorous test of the adaptive coupling law's ability to ensure grid synchronization.

6.3 Non-Linear Dynamic Simulations

The results in section 5 simulate the power grid using the linear Kuramoto dynamics derived in equation (16). The linear derivation assumes that the system is near a synchronized state, which is not always realistic in real world power grids. This assumption simplifies the term representing the power between nodes on the network,

$$w_{i,j}\sin(\varphi_j - \varphi_i), \quad (47)$$

to

$$w_{i,j}(\varphi_j - \varphi_i)\cos(\varphi_j^* - \varphi_i^*), \quad (48)$$

This assumes a constant stiffness between nodes in the system. During large disturbances, when phase deviations exceed leave the linear operating regime, the actual restoration forces weaken. Hence, there are some cases where the linear model would fail to identify divergent phase and frequency events. To improve the robustness of the analysis, we could repeat the simulations from section 5 with nonlinear Kuramoto dynamics.

- more elaborate pinning strategies to minimize usage of non-renewables on the grid
- simulations that can change the intrinsic frequency of nodes
- more nonlinear simulations to more accurately analyze control effectiveness for large disturbances
- if we can extend our results beyond a stable operating point, opens the opportunity for stability augmentation for larger perturbations (Remedial Action Schemes)

7 Summary

TODO

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